

Skein Lasagna Modules Detect Exotic 4-Manifolds

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1 Introduction

The guiding question for this essay is the following question.

Conjecture 1. *When are two homeomorphic (closed) manifolds diffeomorphic?*

In dimensions 1 and 2, it is known classically that any two homeomorphic closed manifolds are diffeomorphic. This continues to be true in dimension 3 due to Moise's theorem [Moi52]. The natural next step is then to ask whether this phenomenon sustains in higher dimensions.

The first counter-examples to Conjecture 1 were given by Milnor, who constructed smooth 7-manifolds which were homeomorphic but not diffeomorphic to S^7 [Mil56]. In this essay we will focus on dimension 4, where the first two landmark results about exotica came from Freedman and Donaldson:

Theorem 1.1 ([Fre82], Theorem 1.5). *Any integral unimodular form Q is realised as the intersection form of some simply connected closed oriented topological 4-manifold M . If Q is even, any two such manifolds are homeomorphic, and if Q is odd, there exists two homeomorphism classes of manifolds realising Q , at most one of which is realised by a smooth manifold.*

Theorem 1.2. [Don83, Theorem 1] *If the intersection form Q_X of a closed simply connected oriented smooth 4-manifold X is positive definite, then in fact it must be equivalent (over $\mathbb{Z}$) to the standard diagonal form.*

These two theorems profoundly reshaped the frontier of 4-manifold topology. For instance, 1.1 implies the 4-dimensional topological Poincaré Conjecture. Theorems 1.1 and 1.2 also imply the existence of non-smoothable topological 4-manifolds, since there exist (many) unimodular definite integer lattices which are not diagonalizable. Mathematicians also soon used Freedman and Donaldson's results to show the existence of an *exotic* $\mathbb{R}^4$; i.e. a 4-manifold homeomorphic but not diffeomorphic to $\mathbb{R}^4$ (see e.g. [Gom83]).

Exotica problems have since been well-studied, and many smooth invariants have been developed to attack them. Notably, to prove Theorem 1.2, Donaldson pioneered the use of gauge theory in studying 4-manifolds, which has become an important source of smooth invariants capable of detecting exotica. These invariants are all analytic in nature, and have lead to a variety of breakthroughs in 4-manifold and knot theory problems. However the current literature has reached some limitations in applying these methods to more difficult exotica problems (e.g. constructing exotic pairs with small b_2), so one may ask if there exists another source for smooth 4-manifold invariants.

One potential breakthrough came via Khovanov homology, a link invariant [Kho00]. Unlike gauge-theory invariants, Khovanov homology can be computed combinatorially. Yet as shown by Rasmussen [Ras10], Khovanov homology can also be used to construct smooth invariants (e.g. Rasmussen's s -invariant). Thus one naturally wondered if one could construct a smooth 4-manifold invariant using Khovanov homology.

Recently, Morrison, Walker, and Wedrich constructed such an invariant: the skein lasagna module [MWW22]. Moreover, in 2024 Ren and Willis showed that the skein lasagna module could indeed distinguish pairs of compact exotic 4-manifolds [RW24], as stated by their theorem below:

Theorem 1.3. [RW24, Theorem 1.1] *Consider the knots K_1, K_2 as shown in Figure 1. Then the knot traces $X_1 := X_{-1}(K_1)$ and $X_2 := X_{-1}(K_2)$ are homeomorphic but not diffeomorphic.*

The exotic pair (X_1, X_2) was first discovered by Akbulut in [Akb91b, Akb91a]. Akbulut proved X_1, X_2 are homeomorphic by performing a sequence of Kirby moves showing a diffeomorphism $\partial X_1 \cong \partial X_2$, which extends to a homeomorphism $X_1 \cong X_2$ by [Boy86, Theorem 0.1, Proposition 0.6]. Hence proving X_1, X_2 are not diffeomorphic will be the main objective of this essay.

This essay will proceed as follows. In Section 2 we introduce Khovanov homology, as well as some related tools (e.g. Lee homology and the s -invariant). In Section 3 we define the skein lasagna module of a smooth 4-manifold. We also present some surgery formulas for the skein lasagna module in Subsection 3.2.

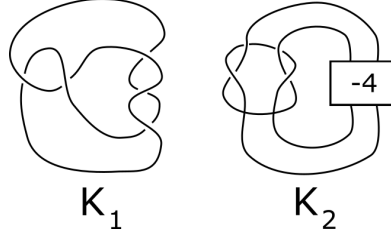


Figure 1: The knots K_1, K_2 used in Theorem 1.3. The knot K_1 is the mirror image of the knot 5_2 (from the Rolfsen table [Rol76, Appendix C]), while the knot K_2 is also the pretzel knot $P(3, -3, -8)$.

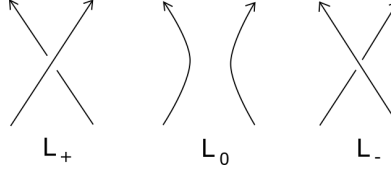


Figure 2: The resolutions of $L_{\pm}$ and L_0 at the specified crossing.

In Section 4 we follow [RW24] and characterize the Lee skein lasagna module of a 4-manifold. In Section 5 we present Ren and Willis' proof of Theorem 1.3. Finally in 6 we discuss a recent Skein-Lasagna-free Khovanov proof of the existence of exotic Mazur manifolds by Nahm [Nah25], and discuss which results Khovanov homology has yet to conquer.

In this paper we use the following conventions. Throughout this essay, *blue font* marks definitions and *green font* marks important remarks. For a manifold M , let $\overline{M}$ denote its orientation reversal, $\nu(M)$ its tubular neighborhood (if M is embedded), and $-M$ its mirror (if M is a link in S^3). We will use ∂ to denote boundary. Also $\#$ (resp. $\natural$) denotes the (resp. boundary) connect sum operation. Moreover, given a framed link $K \subset S^3 = \partial B^4$, we will let its *trace* be the manifold obtained by attaching 2-handles to B^4 along the components of K . Finally, we will denote $I = [0, 1]$.

2 Khovanov Homology

This section follows [BN02], and covers the work of Khovanov in [Kho00].

2.1 Introduction: Jones Polynomial

Khovanov's original motivation for Khovanov homology was to “categorify” the Jones polynomial; i.e. construct a bigraded homology group for each link K in S^3 with normalized Euler characteristic the Jones polynomial [Kho00]. Thus we first define the Jones polynomial to foreshadow our construction of Khovanov homology.

The Jones polynomial $V_L(q)$ is a polynomial invariant of oriented links in S^3 , mainly characterized by the following skein relation. Consider oriented links L_+, L_-, L_0 which are identical aside from their resolutions at a given crossing (see Figure 2). Then their Jones polynomials satisfy [Jon87, (11.1)]

$$(q^{1/2} - q^{-1/2})V_{L_0}(q) = q^{-1}V_{L_+}(q) - qV_{L_-}(q).$$

Given this skein relation along with some starting axioms, one can readily compute the Jones polynomial of any knot. But why is the Jones polynomial a well-defined link invariant? One way to show the well-definedness of the Jones polynomial is to work with its close cousin, the *Kauffman bracket*.

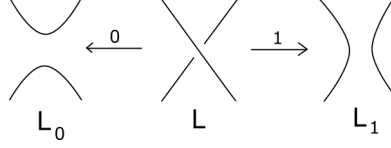


Figure 3: The resolutions of L , L_0 , and L_1 at the specified crossing.

Definition 2.1 ([Kau87]). *Given any unoriented link diagram L , its associated Kauffman bracket $\langle L \rangle$ is a polynomial satisfying the following 3 properties:*

1. *We have $\langle \emptyset \rangle = 1$.*
2. *Given any link diagram L , the split union diagram $U \sqcup L$ satisfies $\langle U \sqcup L \rangle = (q + q^{-1})\langle L \rangle$.*
3. *Given three links L , L_0 , L_1 differing only at one crossing as shown in Figure 3, we have that*

$$\langle L \rangle = \langle L_0 \rangle - q\langle L_1 \rangle. \quad (2.1)$$

One can formulaically compute the Kauffman bracket using the Kauffman skein relation as follows. The first two rules imply that any link diagram D consisting of k (unknot) components and 0 crossings has $\langle D \rangle = (q + q^{-1})^k$. Now given any link diagram L with d crossings, compute $\langle L \rangle$ by drawing all 2^d resolutions of L , and adding the corresponding Kauffman brackets via rule 3 above.

Using the skein relation (2.1) one can show that $\langle L \rangle$ is invariant under Reidemeister II and III moves. However a positive Reidemeister I move multiplies $\langle L \rangle$ by $-q^3$. Now given a link diagram L denote its *writhe* as $w(L) = \#\{\text{positive crossings}\} - \#\{\text{negative crossings}\}$; then $f_L(q) := (-q^3)^{-w(L)}\langle L \rangle$ is an oriented link invariant. In fact, one can show that $V_L(q) = f_L(q^{-1/4})$ [Kau87, Theorem 2.8] (using the respective skein relations), so we may retrieve the Jones polynomial from the Kauffman bracket and vice versa.

In our computation of the Kauffman bracket, each resolution of our diagram was given a power of $(q + q^{-1})$ based on the number of components it had. **To construct Khovanov homology, we will upgrade this to a collection of bigraded free abelian groups.** We will replace our bracket skein relation with a differential between resolutions, and taking the homology of this “cube complex” will give us Khovanov homology.

2.2 Defining Khovanov Homology

2.2.1 Input TQFTs

The previous section outlined our construction of the abelian group associated to our Khovanov chain complex. To construct the differentials, we will need a $1 + 1$ -dimensional *topological quantum field theory* (TQFT), which we will discuss now.

Let **2Cob** denote the category whose objects are oriented closed 1-manifolds (i.e. disjoint unions of circles), and whose morphisms are oriented cobordisms (surfaces) Σ from an “in-boundary” M_0 to an “out-boundary” M_1 . The disjoint union $\sqcup$ makes 2Cob into a monoidal category, and the braid cobordism (swapping two circle components) upgrades 2Cob into a symmetric monoidal category.

Given a (commutative unital) ring R , let **Rmod** denote the (symmetric monoidal under $\otimes$) category whose objects are R -modules and morphisms are R -module maps. Then a **$(1 + 1)$ -dimensional TQFT over R** is a symmetric monoidal functor from 2Cob to Rmod. A $(1 + 1)$ TQFT must associate a disjoint union of n circles to $V^{\otimes n}$ for some R -module V . Moreover, by considering handle decompositions, one may readily show that 2Cob is generated by the braid and identity cobordisms, alongside the 4 cobordisms shown in Figure 4. Thus a $(1 + 1)$ -TQFT is determined by the maps it associates to the 4 cobordisms in Figure 4. In our case, we will set R to be $\mathbb{Z}$ (or $\mathbb{Q}$), and define the following TQFT:

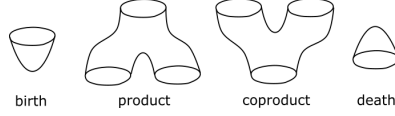


Figure 4: 4 generators for the category 2Cob. The cobordisms are to be read from bottom to top.

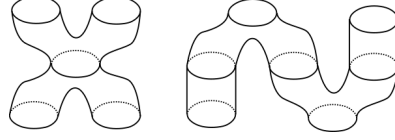


Figure 5: The left cobordism displays $\Delta \circ m$ while the right cobordism displays $(m \otimes 1) \circ (1 \otimes \Delta)$. As shown in the diagram, these two cobordisms are isomorphic.

Definition 2.2 ([BN05], [Kho00], [Lee05]). The [Bar-Natan TQFT](#) is the TQFT over $\mathbb{Z}$ which sends the unknot to $V = \mathbb{Z}[X]/(X^2 - t)$, and induces the following maps:

1. On the birth cobordism it maps $1 \in \mathbb{Z}$ to $1 \in V$;
2. On the death cobordism it gives the map $\epsilon : V \rightarrow \mathbb{Z}$ determined by $\epsilon(1) = 0$, $\epsilon(X) = 1$.
3. On the product cobordism it maps $m : V \otimes V \rightarrow V$ via the usual product of integer polynomials.
4. On the coproduct cobordism it gives the map $\Delta : V \rightarrow V \otimes V$ determined by

$$\begin{aligned}\Delta(1) &= 1 \otimes X + X \otimes 1, \\ \Delta(X) &= X \otimes X + t(1 \otimes 1).\end{aligned}$$

Specializing to $t = 0$ and $t = 1$ (the latter with $\mathbb{Q}$ coefficients) gives us the [Khovanov](#) and [Lee](#) TQFTs.

To show that the Bar-Natan TQFT is well-defined, it suffices to check that several commutativity and associativity laws, as well as the so-called Frobenius law

$$\Delta \circ m = (m \otimes 1) \circ (1 \otimes \Delta). \quad (2.2)$$

Topologically, this amounts to checking that the two cobordisms in Figure 5 induce the same map (which one should expect, since the cobordisms are isomorphic). The underlying idea here is a canonical equivalence between the category of $(1 + 1)$ -TQFTs over any field k , and the category of Frobenius algebras over k [Koc04, Theorem 3.3.2]. But we will not need this theorem.

2.2.2 The Cube Complex

We now define [Khovanov homology](#). This is a bigraded abelian group $\text{Kh}^{i,j}$, where i (resp. j) is called the [homological](#) (resp. [Jones](#) or [quantum](#)) grading. Given a homologically (resp. quantum) graded abelian group $C = \bigoplus_i C_i$, let $C[s]$ (resp. $C\{s\}$) denote C with its homological (resp. quantum) grading shifted by s (explicitly, $C[s]_k = C_{s-k}$). Given a homogeneous element $x \in C$, denote its homological and quantum gradings respectively by $h(x)$, $q(x)$.

Now consider an oriented link diagram D with n crossings. Throughout, let $V = \mathbb{Z}[X]/(X^2)$; give V a bigrading by requiring $q(1) = 1$, $q(X) = -1$, and $h(1) = h(X) = 0$. Following [BN02], we construct the Khovanov homology $\text{Kh}^{i,j}(D)$ of D as follows:

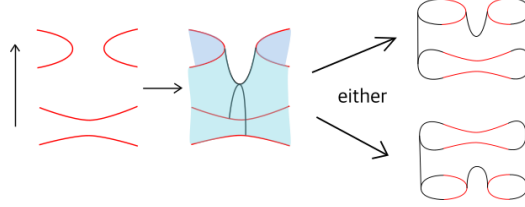


Figure 6: On the left we show how changing a 0 resolution to a 1 resolution gives a saddle cobordism, with the resolved arcs drawn in red. On the right we show how this cobordism can lead to either a product or coproduct cobordism.

1. Pick a labelling of the crossings from 1 to n . For each bit $\pi \in [0, 1]^n$ associate a resolution D_π of D to π by resolving the i th crossing by the $\pi(i)$ resolution, where 0 and 1 resolutions are as specified in Figure 3. Note this resolution is independent of the orientation of D .

2. Each bit π receives a resolution D_π consisting of c_π unknotted circles for some integer c_π . Then associate $C_\pi = V^{\otimes c_\pi}[\pi][\{|\pi|\}]$ to π , where $|\pi|$ denotes the number of 1 entries in π .

This already constructs the bigraded abelian group $\bigoplus_\pi C_\pi$ associated to the Khovanov complex $C^{i,j}(D)$ of D , up to a final grading normalization we will later impose. We now construct the differential.

3. The 2^n bits π are situated in a “hypercube”, where two bits π_0, π_1 differing at exactly one index are connected by a directed edge $\xi : \pi_0 \rightarrow \pi_1$. Denote such edges by bits in $\{0, 1, *\}^n$, where $*$ is only used at the index whose entries differ in π_0 and π_1 . Let $e(\xi)$ denote the number of 1 entries of ξ preceding its $*$ entry.
4. Changing between a 0 and a 1 resolution may be viewed as a saddle cobordism, as shown in Figure 6. After adding in the rest of D_{π_0} and D_{π_1} , we obtain either a product or coproduct cobordism, disjoint union with several identity cobordisms (one for each component not touching the resolved arcs). Applying the Khovanov TQFT to this cobordism gives a map $C_{\pi_0} \rightarrow C_{\pi_1}$, which we denote d_ξ . Note that d_ξ is always $(1, 0)$ -graded due to the $\{|\pi|\}$ grading shift on C_π .
5. Define the differential d on C_π by

$$d = \sum_{\xi} (-1)^{e(\xi)} d_\xi$$

where ξ runs through all edges pointing out of π . Extend d linearly to a differential on all of $\bigoplus_\pi C_\pi$.

6. Let n_+ (resp. n_-) denote the number of positive (resp. negative) crossings in D . Define the Khovanov complex as:

$$C^{i,j}(D) = \left(\left(\bigoplus_{\pi \in [0,1]^n} C_\pi \right) [-n_-] \{n_+ - 2n_-\}, d \right).$$

See [BN02, (3)] for an illustration of the construction of $C^{i,j}(D)$ where D is the usual diagram of the right-handed trefoil. We now need to check this construction actually gives a chain complex.

Theorem 2.1 ([Kho00], Proposition 8). *The differential d on $C^{*,*}(D)$ satisfies $d^2 = 0$.*

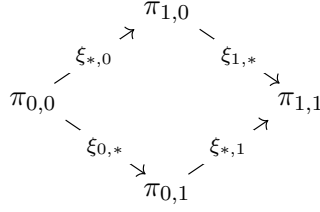


Figure 7: A 2-dimensional face of $[0, 1]^n$. Here $\pi_{a,b}$ has $\pi_{a,b}(i) = a$, $\pi_{a,b}(j) = b$.

Proof. Consider any 2-dimensional face of the hypercube $[0, 1]^n$, whose vertices are a 4-tuplet of bits differing at 2 entries (which we call i and j , where $i < j$). Label the edges as shown in Figure 7. We claim that

$$d_{\xi_{1,*}} \circ d_{\xi_{*,0}} = d_{\xi_{*,1}} \circ d_{\xi_{0,*}}. \quad (2.3)$$

Indeed, the cobordism maps induced on both sides are the compositions of the cobordism maps induced by the two resolution changes at crossings i and j . Composing these cobordisms in either way give the same resulting cobordism; hence the two resulting maps are the same, proving (2.3).

Now observe $e(\xi_{1,*}) + e(\xi_{*,0}) = e(\xi_{*,1}) + e(\xi_{0,*}) + 1$. Thus the restriction of d to $C_{\pi_{0,0}}$ composed with the projection onto $C_{\pi_{1,1}}$ is the zero map. Summing over all squares gives $d^2 = 0$. $\square$

Since $(C^{*,*}(D), d)$ is now a chain complex, we may define $\text{Kh}^{*,*}(D)$ to be its homology. Observe that given link diagrams L, L_0, L_1 which differ as per Figure 3 (where L has n crossings), we have that up to normalizations, the cube complex $C^{*,*}(L)$ has $C^{*,*}(L_0)$ and $C^{*,*}(L_1)$ as $n - 1$ -dimensional faces. Hence up to suitable normalizations, we may view $C^{*,*}(L)$ as the complex (c.f. [BN02, Page 338])

$$C^{*,*}(L) = (0 \rightarrow C^{*,*}(L_0) \rightarrow C^{*,*}(L_1) \rightarrow 0). \quad (2.4)$$

One may view this as a “categorification” of the Kauffman skein relation (2.1). Now, Khovanov homology also clearly satisfies the identities $\text{Kh}(D \sqcup U) \cong \text{Kh}(D) \otimes V$ and $\text{Kh}(\emptyset) \cong \mathbb{Z}$. We can explicitly compare Khovanov homology and the Kauffman bracket as follows. Define the *Poincaré polynomial* of $\text{Kh}(D)$ as $P_D(t, q) := \sum_{i,j} t^i q^j \text{rank}(\text{Kh}^{i,j}(D))$. Then using the above properties, one may show:

Theorem 2.2 ([Kho00], Proposition 27). *For any link L with link diagram D , we have*

$$J(L) = P_D(-1, q).$$

We are now prepared to discuss:

Theorem 2.3 ([Kho00], Theorem 1). *Given any link L with link diagram D , the bigraded isomorphism class of the group $\text{Kh}^{i,j}(D)$ is independent of the choice of D . That is, $\text{Kh}^{i,j}(L)$ is a well-defined invariant of links in S^3 .*

We will just show $\text{Kh}^{i,j}(D)$ is invariant under negative Reidemeister 1 twists. This proof (as well as a proof for other Reidemeister moves) can be found in [BN02, Section 3.5].

Proof. Consider a link diagram D , and let D' be a negative Reidemeister I twist of it. As shown in Figure 8, each resolution D_π corresponds to two resolutions $D'_{\pi,0}, D'_{\pi,1}$, where $D'_{\pi,0}$ is a resolution of D_0 with diagram D_π , and $D'_{\pi,1}$ is a resolution of D_1 with diagram $D_\pi \sqcup U$. For each π , consider the $(1, 2)$ -graded injection $\phi_\pi : C_\pi \rightarrow C_{\pi,1}$ given by tensoring with $1 \in V$ on the U component (where C_π is in the cube complex of

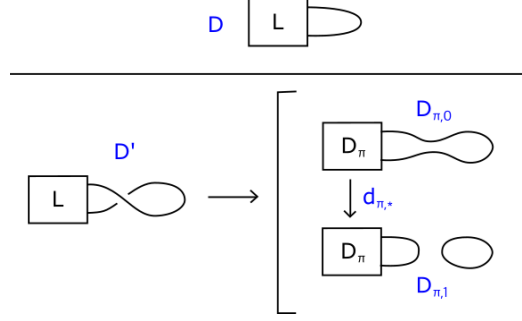


Figure 8: A pictorial description of our proof that Kh is invariant under negative Reidemeister 1 twists. Labels of what each diagram describes are in blue.

D and $C_{\pi,1}$ in the cube complex of D_2). The differential $d_{\pi,*} : C_{\pi,0} \rightarrow C_{\pi,1}$ is a coproduct; hence it is injective and satisfies $\text{im}\phi_\pi \oplus \text{im}d_{\pi,*} = C_{\pi,1}$. By applying ϕ_π to each factor C_π , we obtain an injection

$$\phi : C^{*,*}(D) \rightarrow C^{*,*}(D_2)$$

which is now $(0,0)$ -graded since the new negative intersection shifts the bigrading on $C^{*,*}(D')$ by $(-1, -2)$. On the cokernel of ϕ , we have that the differentials $d_{\pi,*}$ descend to isomorphisms. Thus $H_*(C(D')/C(D)) = 0$, so by the long exact sequence on homology we obtain $\text{Kh}(D) \cong \text{Kh}(D')$ as desired. $\square$

2.3 The Lee Deformation

In our construction of Khovanov homology, the Khovanov TQFT was used three times: 1) in the definition of V , 2) in the definition of d_ξ , and 3) when noting that d_ξ has bigrading $(1,0)$. If we instead use the Lee TQFT, V is left unchanged as an abelian group, but the cobordism maps are changed, so d gains an extra factor which we denote Φ . Explicitly, Φ takes $X \otimes X \rightarrow 1$ in the multiplication map and $X \rightarrow 1 \otimes 1$ in the co-multiplication map, so Φ has bigrading $(1,4)$. Since the cobordism maps are still defined via a TQFT, our proof that $d^2 = 0$ still holds, hence we may define $\text{Lee}(D)$ to be the homology of $(\oplus_\pi C_\pi, d + \Phi)$. In fact this homology group gives another link invariant:

Theorem 2.4 ([Lee05], Section 4.2.2). *Given a link L with link diagram D , the isomorphism class of $\text{Lee}(D)$ does not depend on the diagram D we choose. So the Lee Homology $\text{Lee}(L)$ is an invariant of links in S^3 .*

We will always work with Lee homology over $\mathbb{Q}$ coefficients. Now, $\text{Lee}(L)$ evidently inherits a homological grading. However since d and Φ are not bigraded of the same degree, we will have to more carefully describe the usual quantum grading on $\text{Lee}(L)$. To do this, let $C_j(L)$ denote the subspace of elements in $C^{*,*}(L)$ spanned by elements with quantum grading at least j . This descends to a (decreasing) filtration $\{S_j(L)\}$ on $\text{Lee}(L)$, by setting a class $[x]$ to be in $S_j(L)$ if some representative of it is in $C_j(L)$; note this filtration is of finite length. We then obtain a quantum grading $\text{gr} = \text{gr}_{q,\text{Lee}}$ on $\text{Lee}(L)$ by letting $\text{gr}([x])$ denote the maximal j such that $\text{gr}([x]) \in S_j(L)$ (where we set $\text{gr}([0]) = \infty$ by convention).

Note that since Φ increases quantum grading, it respects the filtration $\{C_j(L)\}$. Thus one may construct the filtration spectral sequence associated to $(C(L), d + \Phi)$, which has the following properties:

Theorem 2.5 ([Ras10], Theorem 2.1). *The spectral sequence associated to the filtered complex $(C(L), d + \Phi)$ has E_2 page given by Khovanov homology $\text{Kh}(L) \otimes \mathbb{Q}$, and converges to (the associated graded of) $\text{Lee}(L)$ on the E_∞ page.*

This spectral sequence will later be a useful tool for comparing $\text{Kh}(L)$ and $\text{Lee}(L)$.

2.3.1 The Structure of Lee Homology

In general the Khovanov homology of a link is quite complicated, which makes it a powerful invariant. For instance, no closed form is known for the Khovanov homologies of torus links! On the other hand, Lee homology is much simpler:

Theorem 2.6 ([Lee05], Theorem 4.2). *For every c -component link L , $\text{Lee}(L) \cong \mathbb{Q}^{2^c}$.*

The main idea behind theorem 2.6 is the following. Let m', Δ' denote the (co)multiplication maps of the Lee TQFT. Perform the base change $a = (1 + X)/2$, $b = (1 - X)/2$. Under this basis, m' and Δ' behave as follows:

$$\begin{aligned} m'(a \otimes a) &= a, & m'(b \otimes b) &= b, & m'(a \otimes b) &= m(b \otimes a) = 0, \\ \Delta'(a) &= 2a \otimes a, & \Delta'(b) &= 2b \otimes b. \end{aligned}$$

Now each orientation $\mathfrak{o}$ of L corresponds to a generator of $\text{Lee}(L)$ as follows. Each crossing of a diagram D of L has a unique resolution which preserves $\mathfrak{o}$. So $\mathfrak{o}$ corresponds to a resolution D_π of D , which consists of a disjoint union of circles in the plane. Label each circle by the number of circles it strictly lies within. Increment this label by 1 if a circle is oriented clockwise. Then assign the generator $a \in V$ to a circle with even label, and $b \in V$ otherwise. Tensoring these generators gives an element of $C_\pi \subset C(L)$, which we denote $x_\mathfrak{o}$.

One can now prove that the elements $\{x_\mathfrak{o}\}$ form a linearly independent set in $\text{Lee}(L)$, thus proving $\dim_{\mathbb{Q}} \text{Lee}(L) \geq 2^c$. An induction argument on the number of crossings then implies that in fact $\dim_{\mathbb{Q}} \text{Lee}(L) = 2^c$ as desired.

2.3.2 The s -invariant

Theorem 2.6 characterizes $\text{Lee}(L) \cong \mathbb{Q}^{2^c}$ but says little about its bigrading. In fact, Rasmussen showed that in the case of a knot K , these bigradings are also well-behaved:

Theorem 2.7 ([Ras10], Proposition 3.3). *For any knot K , $\text{Lee}(K)$ is supported on two bigradings which differ by quantum grading 2 (and have the same homological grading).*

In view of Theorem 2.7, Rasmussen defined:

Definition 2.3 ([Ras10], Definition 3.4). *Given a knot K , its s -invariant $s(K)$ is the arithmetic mean of the quantum gradings on which $\text{Lee}(K)$ is supported.*

This definition extends readily to oriented links: given an orientation $\mathfrak{o}$ on a link L , an extension of the proof of [Ras10, Proposition 3.3] implies that $q(x_\mathfrak{o} + x_{\bar{\mathfrak{o}}})$, $q(x_\mathfrak{o} - x_{\bar{\mathfrak{o}}})$ differ by 2. Thus we may define

$$s(L) = s(L, \mathfrak{o}) := \frac{\text{gr}_{q, \text{Lee}}(x_\mathfrak{o} + x_{\bar{\mathfrak{o}}}) + \text{gr}_{q, \text{Lee}}(x_\mathfrak{o} - x_{\bar{\mathfrak{o}}})}{2},$$

as first done in [BW08, Section 6.1]. Note that we immediately have $s(L) = q(x_\mathfrak{o}) - 1$.

2.4 Cobordism Maps

One important feature of Khovanov homology is that it forms a TQFT of its own! More precisely, consider a smooth cobordism $\Sigma \subset \mathbb{R}^3 \times I$ with boundary components $L \subset \mathbb{R}^3 \times \{0\}$ and $L' \subset \mathbb{R}^3 \times \{1\}$. Then as described in [Kho00, Section 6.3], Σ defines a map $\phi_\Sigma : \text{Kh}(L) \rightarrow \text{Kh}(L')$ as follows:

1. After suitably perturbing Σ (relative to its boundary), decompose Σ into a sequence of link isotopies and handle attachments (performed locally). Thus it suffices to define ϕ_Σ on each of these pieces.
2. As shown earlier, a link isotopy induces an isomorphism of Khovanov homology groups, so define ϕ_Σ on each link isotopy as this isomorphism.
3. A 0-handle attachment corresponds to split unioning an unknot, so define ϕ_Σ on this piece as given by tensoring with $1 \in V$ (i.e. the unit map).
4. A 2-handle attachment corresponds to deleting an unknot, so define ϕ_Σ on this piece as given by evaluation under the counit map on the unknot component (so $x \rightarrow 1$ and $1 \rightarrow 0$ on V).
5. Finally, consider a 1-handle attachment cobordism between two link diagrams D, D' . The 1-handle attachment corresponds to a band-sum along an arc intersecting the two diagrams only at its endpoints. Each resolution π of D corresponds to a resolution π' of D' , and on each such resolution this 1-handle induces a (co)product cobordism, which gives a (co)product map $C_\pi \rightarrow C_{\pi'}$. These maps glue to a map $\psi : C(D) \rightarrow C(D')$. Since ψ and the differentials on $C(D), C(D')$ are both defined via the TQFT, ψ commutes with these differentials. Thus ψ induces a map $\text{Kh}(D) \rightarrow \text{Kh}(D')$.

This construction works analogously in the Lee scenario. In fact the canonical Lee generators interact very nicely with these cobordism maps:

Theorem 2.8 ([Ras10], Proposition 4.1). *Consider an oriented cobordism $\Sigma : L \rightarrow L'$, where the orientation on Σ induces orientations $\mathfrak{o}$ on L , $\mathfrak{o}'$ on L' . Suppose every component of Σ has a boundary component in L . Then the cobordism map $\phi_\Sigma : \text{Lee}(L) \rightarrow \text{Lee}(L')$ maps $[x_\mathfrak{o}]$ to a nonzero multiple of $[x_{\mathfrak{o}'}]$.*

One can prove Theorem 2.8 again by decomposing Σ into elementary cobordisms.

2.4.1 A Concordance Homomorphism

One of the most important results about Lee homology is that the s -invariant is a *smooth concordance homomorphism*. This will follow from the following genus bound, where g_4 denotes the smooth slice genus:

Theorem 2.9 ([Ras10], Theorem 1). *For all knots K we have $|s(K)| \leq 2g_4(K)$*

Proof Sketch. By checking on elementary cobordisms, one can show that the induced map ϕ_Σ of a cobordism Σ has quantum filtration degree $\chi(\Sigma)$ (and homological degree 0). Now consider a slice disk bounding K ; puncture this disk to get a knot cobordism $\Sigma : K \rightarrow U$. Consider the orientation $\mathfrak{o}$ on K such that $x_\mathfrak{o}$ attains the maximum quantum grading on Lee. Pick an orientation $\mathfrak{O}$ on Σ restricting to $\mathfrak{o}$ on K , and suppose $\mathfrak{O}$ restricts to the orientation $\mathfrak{o}'$ on U . Then

$$\begin{aligned} s(K) + 1 = \text{gr}_{q, \text{Lee}}([x_\mathfrak{o}]) &\leq \text{gr}_{q, \text{Lee}}([x_{\mathfrak{o}'}]) - \chi(\Sigma) \leq 1 - \chi(\Sigma), \\ s(K) &\leq 2g(\Sigma) = 2g_4(K). \end{aligned}$$

where for the last inequality in the first line we note that the only quantum gradings on $\text{Lee}(U)$ are ± 1 .

Finally to show $-s(K) \leq 2g_4(K)$, one may construct an isomorphism between the Khovanov complex $C(-K)$ of the mirror of K , and the dual $C(K)^*$ of the Khovanov complex of K (see [Ras10, Proposition 3.9]). This shows $-s(K) = s(-K)$, hence $-s(K) \leq 2g_4(-K) = 2g_4(K)$ as desired. $\square$

Theorem 2.10 ([Ras10], Theorem 2). *The s -invariant gives a homomorphism from the smooth knot concordance group $\mathcal{C}$ to $\mathbb{Z}$.*

Sketch. To show $s(K \# K') = s(K) \# s(K')$, connect-sum K and K' via a band containing a half-twist. Consider the diagrams obtained by resolving the crossing at this half-twist; one gives a diagram for $K \# \overline{K'}$, the other a diagram for $K \sqcup K'$. One can retrieve $s(K \# K') = s(K) \# s(K')$ by considering the resulting exact sequence of Khovanov complexes; see [Ras10, Proposition 3.11] for details.

Now if K, K' are concordant then $K \# -K'$ is slice, so by the genus bound $0 = s(K \# -K') = s(K) - s(K')$. This finishes. $\square$

Theorem 2.10 is very important theoretically, as it gives the first evidence that Khovanov homology is sensitive to the smooth category, and is not just a topological invariant. For example, the s -invariant was invented by Rasmussen to prove the Milnor conjecture (stating that the smooth slice genus of a torus knot coincides with its Seifert genus) [Ras10, Corollary 1], a statement which is false if one works instead with the topological slice genus [BFL18, Theorem 1]. Thus the s -invariant foreshadowed the potential for Khovanov homology to answer further exotica questions in 4-manifold topology.

Remark 2.1. In fact, the s -invariant is enough to show the existence of an exotic $\mathbb{R}^4$. Indeed, consider the following example from [HO08], introduced in the Knot Concordances class. Let T be the right-handed trefoil and let $\text{Wh}_+(T)$ be its positive Whitehead double. We may compute $s(\text{Wh}_+(T)) = 2$, so $\text{Wh}_+(T)$ is not smoothly slice. However since $\text{Wh}_+(T)$ has trivial Alexander polynomial, we have $\text{Wh}_+(T)$ is topologically slice by a result of Freedman [FQ90, Theorem 11.7B]. The trace embedding lemma now implies $X_0(\text{Wh}_+(T))$ embeds topologically but not smoothly in $\mathbb{R}^4$. Take the complement X of the interior of the topological embedding $\varphi : X_0(\text{Wh}_+(T)) \rightarrow \mathbb{R}^4$. Since X is noncompact, a result by Quinn [FQ90, Theorem 8.4] implies that X is smoothable. Since smooth structures on 3-manifolds are unique [Moi52], we may glue the smoothed-out X with $\varphi(X_0(\text{Wh}_+(T)))$ along their boundaries to get an exotic $\mathbb{R}^4$.

Key to this result is the assumption that X is noncompact; otherwise we would have been able to construct an exotic S^4 ! We will still need to do more work to show a compact exotic pair, starting by introducing skein lasagna modules in Section 3.

2.5 Functoriality and Khovanov-Rozansky

So far we have not proven (nor used) the assertion that the Khovanov cobordism maps are independent of the handle decomposition we choose on Σ . In fact this is only true up to sign; see [Jac04] and [Kho06]. One modified homology theory which fixes this sign issue is the *Khovanov-Rozansky homology* KhR (first defined in [KR08]), an invariant of *framed* oriented links in $\mathbb{R}^3$. Khovanov-Rozansky homology has many variants depending on the input lie algebra used: in this essay we will exclusively use the $\mathfrak{gl}_2$ variant, denoted KhR_2 . Note that analogously to Khovanov homology, KhR_2 also has a Lee deformation, which we will denote KhR_{Lee} . Now as worked out in [Bla10] and [ETW18], KhR_2 is functorial over $\mathbb{Z}$ under oriented link cobordisms in $\mathbb{R}^3 \times I$. Most importantly, KhR_2 also coincides with usual Khovanov homology up to the following renormalization [MN22, (6)]:

$$\text{KhR}_2^{i,j}(L) \cong \text{Kh}^{i,-j-w(L)}(-L) \quad (2.5)$$

Here L is a framed oriented link, and its writhe $w(L)$ is its framing number. More formulas relating KhR and Kh are given in Subsubsection 2.5.1.

In our construction of skein lasagna, we will need to consider Khovanov cobordism maps between two links, one in each boundary component of an $S^3 \times I$, where our copy of $S^3 \times I$ is embedded in a general 4-manifold X , and the inner sphere $S^3 \times \{0\}$ bounds a 4-ball in X . To be able to perform this construction we will need much stronger functoriality properties about KhR :

1. We will need to upgrade KhR to a functorial link invariant in *unparameterized* copies of $\mathbb{R}^3$, i.e. manifolds diffeomorphic to $\mathbb{R}^3$. This is worked out carefully in [MWW22, Section 4.1], and may be proven using the known functoriality of KhR over $\mathbb{Z}$.

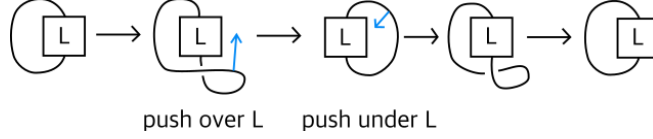


Figure 9: A diagram of the sweep around move. In the second arrow, the arc is pushed over the link L , and in the third arrow, the arc is pushed under the link L .

2. We will then need to extend KhR to a link invariant in unparameterized copies of S^3 , which associates well-defined Khovanov maps to framed link cobordisms in unparameterized copies of $S^3 \times I$. The main new technology needed here is that an isotopy which loops a small arc of a link L around the point of infinity $\infty \in S^3$ (which we would otherwise delete to form $\mathbb{R}^3$) should induce the identity map on $\text{KhR}(L)$. One may alternatively view this as taking a small arc of L and looping it around the rest of L , as shown in Figure 9; this is known as the [sweep-around move](#). The statement that sweep-around moves induce identity maps on KhR is proven in [MWW22, Theorem 1.1], and as shown in [MWW22, Section 4.2] this upgrades KhR to a functorial link invariant in unparameterized S^3 s.

2.5.1 Khovanov Rozansky Formulas

In this section we list a few useful formulas and conventions of Khovanov-Rozansky homology (and its Lee deformation). For more details, see [RW24, Section 2.3]. From this section on, let $h(\cdot)$ and $q(\cdot)$ respectively denote the homological and quantum degrees. Also let $w(L, \mathfrak{o})$ denote the writhe of an oriented link $(L, \mathfrak{o})$, and let $\#L$ denote the number of components of any 1-manifold L . Moreover, we will use an asterisk $\ast \in \{2, \text{Lee}\}$ when an equation holds in both the Khovanov-Rozansky and Lee contexts.

Lemma 2.1. *The Khovanov-Rozansky map $\text{KhR}_2(S)$ induced by a framed oriented link cobordism $S : L_0 \rightarrow L_1$ has bidegree $(0, -\chi(S))$.*

Lemma 2.2 ([RW24], Proposition 2.8). *Let $V = V_{\text{KhR}} := \mathbb{Z}[X]/(X^2)$ and $V_{\text{Lee}} := \mathbb{Q}[X]/(X^2 - 1)$. Then $\text{KhR}_2(U) \cong V_{\text{KhR}}\{-1\}$ and $\text{KhR}_{\text{Lee}}(U) \cong V_{\text{Lee}}\{-1\}$ for U the unknot.*

Note that changing the orientation of a link only changes its Khovanov homology by a grading shift. The same applies for Khovanov-Rozansky and Lee homology:

Lemma 2.3 ([RW24], (10)). *Consider a framed link L , and two orientations $\mathfrak{o}, \mathfrak{o}'$ on L . Then for $\ast \in \{2, \text{Lee}\}$,*

$$\text{KhR}_\ast(L, \mathfrak{o}') \cong \text{KhR}_\ast(L, \mathfrak{o}) \left[\frac{w(L, \mathfrak{o}) - w(L, \mathfrak{o}')}{2} \right] \left\{ \frac{w(L, \mathfrak{o}') - w(L, \mathfrak{o})}{2} \right\}. \quad (2.6)$$

From now on, given a link cobordism $\Sigma : L_0 \rightarrow L_1$, define its [normalized Euler characteristic](#) $n(\Sigma)$ as $n(\Sigma) := (-\chi(\Sigma) + \#L_1 - \#L_0)/2$

Lemma 2.4 ([RW24], Proposition 2.8). *Consider a framed oriented link $L \subset S^3$, and an alternative orientation $\mathfrak{o}$ on L . Let $x_{\mathfrak{o}}$ be the generator of $\text{Lee}(L)$ corresponding to $\mathfrak{o}$. Then:*

1. $x_{\mathfrak{o}}$ is homologically homogeneous with $h(x_{\mathfrak{o}}) = (w(L, \mathfrak{o}) - w(L))/2$;
2. A framed oriented link cobordism $\Sigma : L_0 \rightarrow L_1$ induces the Lee cobordism map

$$\text{KhR}_{\text{Lee}}(\Sigma)(x_{\mathfrak{o}}) = 2^{n(\Sigma)} \sum_{\mathfrak{D}|_{L_0}=\mathfrak{o}} (-1)^{n(\Sigma-(\mathfrak{D}))} x_{\mathfrak{D}|_{L_1}} \quad (2.7)$$

where $\mathfrak{D}$ ranges over all orientations on Σ which align with $\mathfrak{o}$ on L , and $\Sigma-(\mathfrak{D})$ consists of all components of Σ whose orientations differ with $\mathfrak{D}$.

3 Skein Lasagna

3.1 The Definition of Skein Lasagna

We now define the (Khovanov-Rozansky) skein lasagna module associated to any 4-manifold, following the presentation of [RW24, Section 2.1] inspired by Sungkyung Kang. Throughout let a *pair* (X, L) denote a smooth oriented 4-manifold X (possibly with boundary) along with a framed oriented link $L \subset \partial X$. For a link L' embedded inside an arbitrary 3-manifold Y diffeomorphic to S^3 , let $\text{KhR}_2(Y, L')$ denote the Khovanov-Rozansky homology of L' in Y as defined in Subsection 2.5.

Definition 3.1. A *skein* in (X, L) consists of:

1. A finite collection of disjoint embedded 4-balls $\{B_i\}$ in $\text{int} X$;
2. A framed oriented surface Σ properly embedded in $X - \cup_i B_i$, such that $\Sigma \cap \partial X = L$, and for each i , $\Sigma \cap \partial B_i$ is a link L_i .

The skeins of a pair (X, L) form the objects of a category $\mathcal{C}(X, L)$. The morphisms of this category are given by compositions of the following two operations:

1. We have a morphism $\Sigma \rightarrow \Sigma'$ if we may retrieve Σ' from Σ via an isotopy fixing L and the input balls. Call such isotopies *skein isotopies*.
2. Given a skein Σ_1 , consider one of its input balls B . Suppose $\Sigma_1 \cap B = L_1$. Then replacing B with a skein in (B, L_1) (and smoothing out the resulting surface along L_1) gives another skein Σ_0 . In this scenario we also set there to be a morphism $\Sigma_0 \rightarrow \Sigma_1$. We will call this operation the *ball encloement* morphism; see Figure 10 for a diagram.

Note that two skeins Σ, Σ' related by an isotopy fixing L but not the input balls are related by a sequence of morphisms of the two forms above. Now, let $\mathbb{Z}\text{mod}$ denote the category of $\mathbb{Z}$ -modules. Define the *Khovanov Rozansky lasagna filling functor* $Z = Z_{\text{KhR}_2} : \mathcal{C}(X, L) \rightarrow \mathbb{Z}\text{mod}$ as follows. Given a skein Σ with input balls B_i and links L_i , set

$$Z(\Sigma) := \otimes_i \text{KhR}_2(\partial B_i, L_i).$$

Define a *lasagna filling* F to be a skein Σ along with a choice of element in $Z(\Sigma)$.

Now consider a 4-manifold X homeomorphic to B^4 , and a link $L \subset \partial X$. We may interpret any skein Σ on (X, L) as a link cobordism by picking a topological tree T with endpoints on $\{\partial B_i\}$ disjoint from Σ , and then connecting the balls B_i into one ball $B' = (\cup_i B_i) \cup \nu(T)$ for some choice of tubular neighborhood $\nu(T)$. This link cobordism gives a Khovanov map $\text{KhR}_{2, \Sigma}$, which is independent of the choice of $\nu(T)$ and T (due to the sweep-around move; see [MWW22, Section 5.2]). Thus define the functor Z on skein morphisms as follows. Let Z map isotopies of skeins to identity maps. Moreover, consider a skein Σ_0 obtained from Σ_1 by replacing an input ball B_1 with a skein Σ . Then define $Z(\Sigma_0 \rightarrow \Sigma_1)$ as $\text{KhR}_2(\Sigma)$ on the links of Σ_0 inside B_1 , and the identity map on links of Σ_0 outside B_1 .

Definition 3.2 ([MWW22], Definition 5.4). Given a pair (X, L) , the *KhR₂ skein lasagna module* $\mathcal{S}_0^2(X; L)$ of (X, L) is the colimit of its lasagna filling functor $Z : \mathcal{C}(X, L) \rightarrow \mathbb{Z}\text{mod}$. Explicitly,

$$\mathcal{S}_0^2(X; L) := \mathbb{Z}[\text{Lasagna fillings on } X \text{ with boundary } L] / \sim$$

where $\sim$ is defined as the transitive and linear closure of the following relations:

1. Lasagna fillings are taken to be multilinear in each of their input elements v_i ;

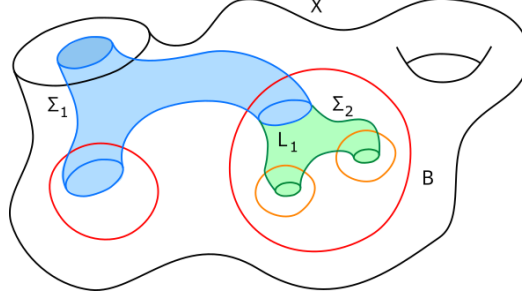


Figure 10: The ball enclosure morphism. Here $\Sigma_0 = \Sigma_1 \cup \Sigma_2$.

2. Given a lasagna filling $F_0 = (\Sigma_0, v_0)$ and a skein morphism $\Sigma_0 \rightarrow \Sigma_1$, set $F_0 \sim (\Sigma_1, Z(\Sigma_0 \rightarrow \Sigma_1)v_0)$.

As the above suggests, it is useful to think of the skein lasagna module as the colimit of the space of lasagna fillings $Z(\Sigma)$ as one takes the input balls of Σ to be “very big”. For instance, any skein Σ admits a morphism to a skein with only one input ball, by enclosing all input balls of Σ in one embedded ball. Moreover, in the case of $X = B^4 \subset \mathbb{R}$, given any skein Σ on (X, L) we can always pick some $\epsilon > 0$ such that outside of $(1 - \epsilon)B^4 \subset B^4$, Σ is isotopic to $L \times I$. This proves (see [MWW22, Example 5.6])

$$\mathcal{S}_0^2(B^4; L) \cong \text{KhR}_2(S^3, L). \quad (3.1)$$

When $L = \emptyset$, we will denote $\mathcal{S}_0^2(X) := \mathcal{S}_0^2(X; \emptyset)$. Also, the superscript 2 in $\mathcal{S}_0^2(X; L)$ indicates that we are working with KhR_2 . The subscript 0 comes from the fact that we may define more abelian groups $\mathcal{S}_i^2(X; L)$ for $i > 0$ via Morrison and Walker’s blob homology construction [MW11]. We will not use this construction, so we will only work with $\mathcal{S}_0^2$ from now on.

Letting $\deg$ denote bidegree, we assign F the bidegree

$$\deg(F) := \sum_i \deg(v_i) + (0, -\chi(\Sigma)).$$

Since a link cobordism Σ_0 induces a KhR map of bidegree $(0, -\chi(\Sigma_0))$, this bidegree is well-defined on each class $[F]$, and hence descends to a bigrading on $\mathcal{S}_0^2(X; L)$. We will let $\mathcal{S}_{0,h,q}^2(X; L)$ denote the (h, q) -bigraded part of $\mathcal{S}_0^2(X; L)$.

As discussed in [RW24, Section 2.1], this construction of the skein lasagna module works for any link homology theory satisfying sufficiently nice properties. Notably we may define the skein lasagna module of (X, L) associated to KhR_{Lee} , which we will denote $\mathcal{S}_0^{\text{Lee}}(X; L)$.

3.1.1 A Homology Decomposition of the Skein Lasagna Module

We have an isomorphism

$$H_2(X, L \cup (\cup_i B_i); \mathbb{Z}) \cong H_2(X, L; \mathbb{Z})$$

Under this isomorphism we may assign any skein Σ or lasagna filling F a homology class in $H_2(X, L; \mathbb{Z})$. Moreover, under the boundary map $\partial : H_2(X, L; \mathbb{Z}) \rightarrow H_1(L)$ we have $\partial([F]) = [L]$, so $[F]$ always lives in $H_2^L(X) := \partial^{-1}([L]) \subset H_2(X, L)$. Note that $[F] \in H_2^L(X)$ is clearly invariant under skein isotopies and ball enclosure morphisms, so the homology class $[[F]] \in H_2^L(X)$ of the class $[F]$ in $\mathcal{S}_0^2(X; L)$ is well-defined.

We may also define an intersection product on skeins as follows. Glue 2-handles to X along our framed link L to obtain a manifold $X(L)$. A skein Σ in (X, L) gives a 2-chain in (X, L) after contracting its input balls, which gives a 2-chain on $X(L)$ after gluing the core of the 2-handle to Σ . Then define the product $\cdot$

via the intersection form on $X(L)$. Alternatively, we may calculate $\Sigma_1 \cdot \Sigma_2$ by pushing Σ_2 off L along its framing and then counting transverse intersections, where we isotope the input balls to be distinct.

Now, define $\mathcal{S}_0^2(X; L, \alpha)$ to be the $\mathbb{Z}$ -span of the classes of lasagna fillings F with $[F] = \alpha \in H_2(X, L; \mathbb{Z})$. This gives us a decomposition

$$\mathcal{S}_0^2(X; L) = \bigoplus_{\alpha \in H_2^L(X)} \mathcal{S}_0^2(X; L, \alpha).$$

It turns out that the above decomposition completely determines which skeins (not lasagna fillings!) are related by morphisms. That is, let $\Omega_{\geq 0}(X; L)$ denote the set of connected components of skeins on (X, L) (where skeins are connected via morphisms).

Theorem 3.1 ([RW24], Theorem 2.13). *The homology map*

$$\begin{aligned} \Omega_{\geq 0}(X; L) &\rightarrow H_2^L(X) \\ \Sigma &\rightarrow [\Sigma] \end{aligned} \tag{3.2}$$

is an isomorphism.

Proof. We show that (3.2) is surjective as follows: any class $\alpha \in H_2^L(X)$ is realised by some immersed surface in X bounding L which generically self-intersects transversely at a finite collection of points. After adding input balls, we may realise α by some embedded surface Σ ; adding further input balls then trivializes the normal bundle of Σ , allowing us to add a framing.

To show (3.2) is injective, we consider the following equivalence relation on $\mathcal{C}(X, L)$. Call two skeins $\Sigma_0, \Sigma_1 \in \mathcal{C}(X, L)$ with input balls $\{B_{i,0}\}, \{B_{i,1}\}$ *framed cobordant* if there exists:

1. a 1-complex Γ with tubular neighborhood $\nu(\Gamma)$ satisfying $\nu(\Gamma) \cap (X \times \{k\}) = \cup_i (\text{int } B_{i,k} \times \{k\})$ for $k \in \{0, 1\}$, and;
2. a properly embedded framed oriented 3-manifold $Y \subset (X \times I) \setminus \nu(\Gamma)$ with $Y \cap (X \times \{k\}) = \Sigma_k \times \{k\}$.

Here think of Γ as a cobordism between input balls and Y as a cobordism between Σ_0 and Σ_1 outside these balls. Then [RW24, Lemma 2.14] states that two skeins lie in the same connected component of $\Omega_{\geq 0}(X; L)$ iff they are framed cobordant. Ren and Willis' proof uses Morse theory to decompose any framed cobordism into a collection of elementary moves, each of which decomposes as a sequence of skein morphisms.

Using the notion of framed cobordism, one can readily show the homology map (3.2) is injective, and hence an isomorphism. Indeed, given any two skeins Σ_0, Σ_1 , enclose their input balls into one large ball whose complement is $X' \subset X$. If $[\Sigma_0] = [\Sigma_1]$, then Σ_0, Σ_1 cobound a properly immersed 3-manifold $Y \rightarrow X' \times I$, whose self-intersections and singularities generically can be contained in the neighborhood of a 1-complex T . After deleting $\nu(T)$ we obtain an embedded 3-manifold, and after deleting more balls intersecting Y we may extend the framings of Σ_0, Σ_1 to Y , retrieving our desired framed cobordism. $\square$

Since any two homologous skeins are related by a sequence of morphisms, one can view $\mathcal{S}_0^2(X; L, \alpha)$ as a measure of how much information is transmitted or lost under the corresponding Khovanov-Rozansky maps. For instance, in the proof of the 2-handle formula we will isotope then ball-enclose our lasagna fillings to be within a certain family of lasagna fillings. The Khovanov-Rozansky maps between these fillings lose a fair bit of information, so the resulting Khovanov-Rozansky skein lasagna modules can be rather large (e.g. $\mathcal{S}_0^2(S^2 \times D^2, \emptyset) \cong \mathbb{Z}[x, y^{\pm 1}]$ [MN22, Theorem 1.2]). However, since Lee cobordism maps are very well-behaved, we will be able to characterize $\mathcal{S}_0^{\text{Lee}}(X; L)$ purely using homological data about the skeins inside. This simple form for $\mathcal{S}_0^{\text{Lee}}$ will allow us to extend many useful definitions and properties (e.g. the s -invariant) to Lee skein lasagna.

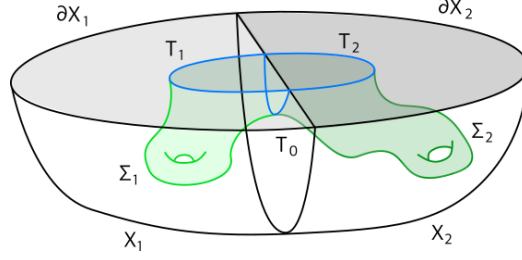


Figure 11: As shown, Σ_1 and Σ_2 glue to a skein on (X, L) .

3.2 Surgery Formulas

We now give many useful tools for computing $\mathcal{S}_0^2(X; L)$. Many of these tools are originally from [MN22], where Manolescu and Neithalath gave formulas for computing $\mathcal{S}_0^2(X; L)$ given a sufficiently nice handle decomposition of X (e.g. with no 1-handles). Note the following theorems all generalize readily to the Lee setting.

Before we present these handle attachment formulas, we first demonstrate that skein lasagna behaves nicely under the following gluing setup. Consider a pair (X, L) and a separating properly embedded oriented 3-submanifold $Y \subset X$, where Y transversely intersects L at some finite collection of points P . Y separates X into two manifolds X_1, X_2 , and L into two *tangles* $T_1 \subset \partial X_1, T_2 \subset \partial X_2$ with $\partial T_1 = \partial T_2 = P$. Now consider a tangle $T_0 \subset Y$, such that $\partial T_0 = P$, and the framing on T_0 restricts on P to the same framing induced by L . Then a skein Σ_1 on $(X_1, T_1 \cup T_0)$ and a skein Σ_2 on $(X_2, -T_0 \cup T_2)$ glue to a skein on (X, L) , as shown in Figure 11. This induces a map:

$$\mathcal{S}_0^2(X_1; T_1 \cup T_0) \otimes \mathcal{S}_0^2(X_2; -T_0 \cup T_2) \rightarrow \mathcal{S}_0^2(X; L). \quad (3.3)$$

Theorem 3.2 ([RW24], Proposition 2.5). *The gluing map (3.3) gives rise to a surjection*

$$\bigoplus_{\substack{T_0 \subset Y \\ \partial T_0 = P}} \mathcal{S}_0^2(X_1; T_1 \cup T_0) \otimes \mathcal{S}_0^2(X_2; -T_0 \cup T_2) \rightarrow \mathcal{S}_0^2(X; L).$$

Proof. We can isotope any skein on (X, L) to intersect Y transversely. □

The following theorem now describe 3 and 4-handle attachments.

Theorem 3.3 ([MN22], Proposition 2.1). *Consider a pair (X, L) . Suppose we attach a k -handle to X away from L to obtain a pair (X', L) . Then this attachment gives a map*

$$\mathcal{S}_0^2(X, L) \rightarrow \mathcal{S}_0^2(X', L) \quad (3.4)$$

which is surjective if $k = 3$, and an isomorphism if $k = 4$.

Proof. The given map (3.4) is induced by inclusion of skeins. Now by definition a k -handle is diffeomorphic to $D^k \times D^{4-k}$, and its cocore $\{0\} \times D^{4-k}$ is an $4 - k$ -dimensional submanifold. Thus we can isotope any skein to transversely intersect this cocore, i.e. be disjoint from it. We may then radially push Σ out of the handle, implying that (3.4) is surjective. In the case $k = 4$, any two equivalent skeins in X' are related by a sequence of skein isotopies and ball enclosure morphisms, both of which can be isotoped and pushed out of the handle. This upgrades (3.4) into an isomorphism. □

Skein lasagna modules also behave nicely under (boundary) connect sums.

Theorem 3.4. [MN22, Theorem 1.4, Corollary 7.3] Consider two pairs $(X_1, L_1), (X_2, L_2)$. Consider $X \in \{X_1 \natural X_2, X_1 \# X_2, X_1 \sqcup X_2\}$, where the (boundary) connect sum is performed away from L_1 and L_2 . Then

$$\mathcal{S}_0^2(X; L_1 \cup L_2; \mathbb{Q}) \cong \mathcal{S}_0^2(X_1; L_1; \mathbb{Q}) \otimes \mathcal{S}_0^2(X_2; L_2; \mathbb{Q}), \quad (3.5)$$

where $\mathcal{S}_0^2(X_0; L_0; \mathbb{Q}) \cong \mathcal{S}_0^2(X_0; L_0) \otimes \mathbb{Q}$ denotes the skein lasagna module taken with $\mathbb{Q}$ coefficients.

The case $X = X_1 \sqcup X_2$ is immediate, and the case $X = X_1 \# X_2$ can be deduced by from the $X = X_1 \natural X_2$ by composing with several 4-handle addition / deletions. In the case $X = X_1 \natural X_2$ where $\partial X_1, \partial X_2$ are connected by some $B^3 \times I$, the idea is to stretch a skein until its given on $B^3 \times I$ by some $L \times I$, then to cut this $L \times I$ in the middle using a new input ball. The two remaining components of the skein, union this new ball, give skeins for (X_1, L_1) and (X_2, L_2) . Using this one can construct the map (3.5).

One can view the boundary sum as attaching a 1-handle connecting two disjoint connected components. A general formula for 1-handle attachments exists (see [MWW23, Theorem 1.3]), but is too complicated and not necessary for our purposes. Interested readers may also find a more precise analysis of the 3-handle attachment map in [MWW23, Theorem 3.7].

3.2.1 The 2-Handle Attachment Formula

The 2-handle attachment formula will reduce the study of skein lasagna modules to the study of the KhR_2 of cabled links. These cabled links arise as the intersection of our skein with the attaching regions of the 2-handles, and are equipped with a braid group action, which we first discuss now. Given a framed link $K \subset S^3$ with components $K_1, \dots, K_n$, and two vectors $k_{\pm} \in \mathbb{Z}_{\geq 0}^n$, let $K(k^+, k^-)$ denote the cabled link obtained by replacing K_i with k_i^+ positively oriented and k_i^- negatively oriented parallel (as determined by the framing) strands in a tubular neighborhood $\nu(K) = \cup_i \nu(K_i)$. Let B_m denote the braid group on m strands. For any integers m, m' , let $B_{m, m'}$ denote the subgroup of $B_{m+m'}$ which sends the first m strands to each other. By identifying each parallel pushoff of K_i with its intersection with a meridional disk in $\nu(K_i)$, we obtain a natural action $\beta_i : B_{k_i^+, k_i^-} \rightarrow \text{Aut}(\text{KhR}_2(K(k^+, k^-)))$ given by isotoping the positive and negative strands of K_i inside νK_i .

We also have Khovanov-Rozansky maps between different $K(k^+, k^-)$, defined as follows. Let $e_i \in \mathbb{Z}^n$ denote the i^{th} standard basis vector. Then augmenting $k^{\pm} \mapsto k^{\pm} + e_i$ adds one positive and negative copy of K_i , which bound an annulus. Puncturing the annulus gives a degree $(0, 1)$ cobordism map

$$\psi_i : \text{KhR}_2(K(k^+, k^-)) \otimes \text{KhR}_2(U) \cong \text{KhR}_2(K(k^+, k^-) \sqcup U) \rightarrow \text{KhR}_2(K(k^+ + e_i, k^- + e_i)).$$

We will let ψ_i^k for $k \in \{0, 1\}$ denote the (degree $(0, 2k)$) map $\text{KhR}_2(K(k^+, k^-)) \rightarrow \text{KhR}_2(K(k^+ + e_i, k^- + e_i))$ obtained by plugging in $x^k \in \text{KhR}_2(U) \cong \mathbb{Z}[X]/(X^2)$. The maps ψ_i^k technically depend on which two oppositely oriented strands are chosen, but we will soon quotient out the B_{k^+, k^-} action, making the ψ_i^k well-defined. We will also call the maps ψ_i (resp. ψ_i^1) (*resp. dotted*) *annular creation maps* (where “dotted” is in the sense of foam notation).

Theorem 3.5 ([MN22] Theorem 1.1, [RW24] Proposition 2.10). Consider the trace $X = X(K)$ of a framed link $K = K_1 \cup \dots \cup K_n \subset S^3$, and consider a link $L \subset S^3$ disjoint from K . Identify any $\alpha \in H_2^L(X; \mathbb{Z})$ with a vector in $\mathbb{Z}^n$ by counting its algebraic intersections with the cocores of the n handles. Denote $\alpha_i^+ = \max(\alpha_i, 0)$, $\alpha_i^- = \min(\alpha_i, 0)$, and $|\alpha| = \sum_i |\alpha_i|$. Then $\mathcal{S}_0^2(X; L, \alpha)$ is isomorphic to the cabled Khovanov-Rozansky homology

$$\underline{\text{KhR}}_{2, \alpha}(K) := \left(\bigoplus_{r \in \mathbb{Z}_{\geq 0}^n} \text{KhR}_2(K(r + \alpha^+, r - \alpha^-) \cup L) \{-|\alpha| - 2|r|\} \right) / \sim \quad (3.6)$$

where $\sim$ is defined as the transitive and linear closure of the relations

$$\beta_i(\sigma)v \sim v, \quad \psi_i^0(v) \sim 0, \quad \psi_i^1(v) \sim v$$

for all $\sigma \in B_{r_i+\alpha_+, r_i-\alpha_-}$ and $v \in \text{KhR}_2(K(\alpha^+ + r, \alpha^- + r) \cup L)$ across all $r \in \mathbb{Z}_{\geq 0}^n$.

In fact the braid group $B_{k_i^+, k_i^-}$ action on a cabled link $\text{KhR}_2(K(k^+, k^-))$ extends to a $B_{k_i^+ + k_i^-}$ action, which descends to a symmetric group $S_{k_i^+ + k_i^-}$ action as follows. Consider the generator σ_j of $B_{k_i^+ + k_i^-}$ swapping the j^{th} and $(j+1)^{\text{th}}$ strands, and let $\mathfrak{o}$ be an orientation on $K(k^+, k^-)$ where the j^{th} and $(j+1)^{\text{th}}$ strands of the cable around K_i have opposite orientations. Note that σ_j does not preserve the orientation $\mathfrak{o}$, but changing the orientation on $K(k^+, k^-)$ only affects $\text{KhR}_2(K(k^+, k^-))$ by a bidegree shift. Since σ_j does not affect the total number of positively oriented strands, by (2.6) σ_j nevertheless induces a bigraded automorphism on $\text{KhR}_2(K(k^+, k^-), \mathfrak{o})$, hence also $\text{KhR}_2(K(k^+, k^-))$. We then have the following theorem:

Theorem 3.6 ([GLW18], Proposition 9). *In the above setup, $\beta_i(\sigma_j) = \beta_i(\sigma_j^{-1})$. Thus the $B_{k_i^+ + k_i^-}$ action descends to a symmetric group action $\tilde{\beta}_i : S_{k_i^+ + k_i^-} \rightarrow \text{Aut}(\text{KhR}_2(K(k_i^+, k_i^-)))$.*

In fact, this allows us to simplify the relations in Theorem 3.5 as follows.

Theorem 3.7 ([MN22], Proposition 3.8). *Work over $\mathbb{Q}$. Then the relation $\sim$ defined in Theorem 3.5 $\text{KhR}_2(K(r + \alpha^+, r - \alpha^-) \cup L)\{-|\alpha| - 2|r|\}$ is equivalent to the relation $\sim'$ given as the transitive and linear closure of the relations*

$$\tilde{\beta}_i(\sigma)v \sim v, \quad \psi_i^1(v) \sim v$$

for all $\sigma \in S_{2r_i+|\alpha|}$ and $v \in \text{KhR}_2(K(\alpha^+ + r, \alpha^- + r) \cup L)$ across all $r \in \mathbb{Z}_{\geq 0}^n$.

We omit the proof of this theorem. Note that Theorem 3.7 allows us to rephrase cabled Khovanov-Rozansky homology as a colimit as follows. Let $S(r) = \prod_i S_{2r_i+|\alpha_i|}$; then we have

$$\text{KhR}_{2,\alpha}(K) = \text{colim}_{r \in \mathbb{Z}_{\geq 0}^n} \text{KhR}_2(K(r + \alpha^+, r - \alpha^-) \cup L)^{S(r)}\{-|\alpha| - 2|r|\} \quad (3.7)$$

where the superscript $S(r)$ denotes the subspace of $S(r)$ -invariant elements, and the maps between the Khovanov-Rozansky groups are given by the ψ_i^1 maps. This colimit form will be useful to us later in Subsection 4.2. When using Theorem 3.5 beyond this section, we will use the relation $\sim'$ of Theorem 3.7. However in the following proof of 3.5 we will still use the original relation $\sim$.

Proof of Theorem 3.5. Similar to the proof of the 3 and 4 handle formulas, the main idea of this proof is to push as much of any given skein into B^4 as possible. Generically, a skein intersects each cocore transversely at finitely many points: pushing the rest of the skein into B^4 then enclosing all input balls in $(1 - \epsilon)B^4$ for some $\epsilon > 0$ leaves an identity cobordism on L as well as a collection of disks for each K_i . Their boundaries form cables around K_i , hence the given formula. We will formalize this idea later when constructing Φ^{-1} .

We are given a link trace $X(K)$, where the 2-handles are attached along $K \subset S^3 = \partial B^4$. View B^4 as the closed unit ball in $\mathbb{R}^4$ (so we may define $(1 - \epsilon)B^4 \subset B^4$ for any small $\epsilon > 0$). First we define a map

$$\tilde{\Phi} : \bigoplus_{r \in \mathbb{Z}_{\geq 0}^n} \text{KhR}_2(K(r + \alpha^+, r - \alpha^-) \cup L)\{-|\alpha| - 2|r|\} \rightarrow \mathcal{S}_0^2(X; L; \alpha)$$

as follows. For any $r \in \mathbb{Z}_{\geq 0}^n$, map $v \in \text{KhR}_2(K(r + \alpha^+, r - \alpha^-) \cup L)\{-|\alpha| - 2|r|\}$ to the class of the lasagna filling F_v with skein Σ_v , satisfying:

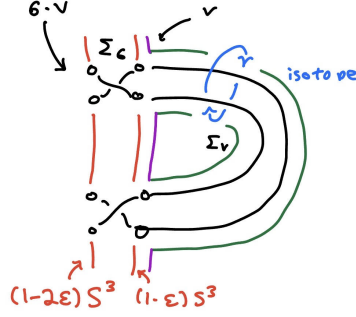


Figure 12: A schematic diagram showing the invariance of $\tilde{\Phi}$ under the $B_{r_i+\alpha_i^+, r_i-\alpha^-}$ action. The cobordism Σ_σ runs right to left; its reverse $\Sigma_{\sigma^{-1}}$ runs left to right. The blue isotopy takes $\Sigma_v \cup \Sigma_v$ to $\Sigma_{\sigma \cdot v}$.

1. Σ_v has one input ball $(1-\epsilon)B^4$, where $\epsilon > 0$ is small;
2. $\Sigma_v \cap \partial(1-\epsilon)B^4 = K(r+\alpha^+, r-\alpha^-) \cup L$, and the decoration of F_v on $K(r+\alpha^+, r-\alpha^-) \cup L$ is v ;
3. For each K_i , Σ_v has $2r_i + |\alpha|$ components, given by $r_i + \alpha_i^+$ positively oriented and $r_i + \alpha_i^-$ negatively oriented disks parallel to the core of the 2-handle bounding K_i . These disks are uniquely framed since they are contractible. Σ_v also has $L \times [1-\epsilon, 1]$ as an extra component.

This construction is clearly independent of $\epsilon > 0$, so $\tilde{\Phi}$ is well-defined. We now show $\tilde{\Phi}$ descends to a map Φ on $\bigoplus_{r \in \mathbb{Z}_{\geq 0}^n} \text{KhR}_2(K(r+\alpha^+, r-\alpha^-) \cup L) \{-|\alpha| - 2|r|\} / \sim$ as follows:

1. Any element $\sigma \in B_{r_i+\alpha_i^+, r_i-\alpha^-}$ induces an isotopy from $K(r+\alpha^+, r-\alpha^-) \cup L$ to itself, which may also be viewed as a link cobordism Σ_σ from $K(r+\alpha^+, r-\alpha^-)$ to itself. Given the skein Σ_v of L_v , replacing the input ball $(1-\epsilon)B^4$ with the link cobordism Σ_σ , along with the smaller input ball $(1-2\epsilon)B^4$ decorated on its boundary with $\sigma \cdot v$ gives us a lasagna filling with skein $\Sigma_v \cup \Sigma_\sigma$ which is isotopic to $\Sigma_{\sigma \cdot v}$, as shown in Figure 12. Σ_σ now induces the reverse cobordism $\Sigma_{\sigma^{-1}}$ outwards which maps $\sigma \cdot v$ to v in $\text{KhR}_2(K(r+\alpha^+, r-\alpha^-) \cup L)$, so we retrieve a ball enclosure morphism from $L_{\sigma \cdot v}$ to L_v . Hence $\tilde{\Phi}$ is invariant under the $B_{r_i+\alpha_i^+, r_i-\alpha^-}$ -action.
2. Consider any $r \in \mathbb{Z}_{\geq 0}^n$ and $v \in \text{KhR}_2(K(r+\alpha^+, r-\alpha^-) \cup L)$. Pick two oppositely oriented strands $K_i^\pm$ of K_i in $\text{KhR}_2(K(r+\alpha^+ + e_i, r-\alpha^- + e_i) \cup L)$ bounded by disks $C_i^\pm$. First, given the lasagna filling L_v with skein Σ_v , replace the input ball $(1-\epsilon)B^4$ of Σ_v with $(1-2\epsilon)B^4$, where the link cobordism Σ is the identity cobordism on $K(r+\alpha^+, r-\alpha^-) \cup L$ union a counit with boundary $U \subset (1-2\epsilon)B^4$. Denote $\Sigma_v \cup \Sigma = \Sigma'_v$; then in $S_0^2(X; L)$, for $k \in \{0, 1\}$ we have the equivalences

$$[(\Sigma'_v, v \otimes X^k)] = [(\Sigma_v, \text{KhR}_{2, \Sigma}(v \otimes X^k))] = \begin{cases} 0 & k = 0 \\ [(\Sigma_v, v)] & k = 1 \end{cases} \quad (3.8)$$

by the definition of the counit map. On the other hand, let

$$Z : K(r+\alpha^+, r-\alpha^-) \cup L \cup U \rightarrow K(r+\alpha^+ + e_i, r-\alpha^- + e_i) \cup L$$

be the link cobordism given by the identity on $K(r+\alpha^+, r-\alpha^-) \cup L$ union the comultiplication cobordism connecting U and $K_i^+ \cup K_i^-$ as before. We claim that $Z \cup C_i^+ \cup C_i^-$ is isotopic rel ∂ to Σ , via the following isotopy (which we will hence denote the *push-in move*). Push Σ into the 2-handle;

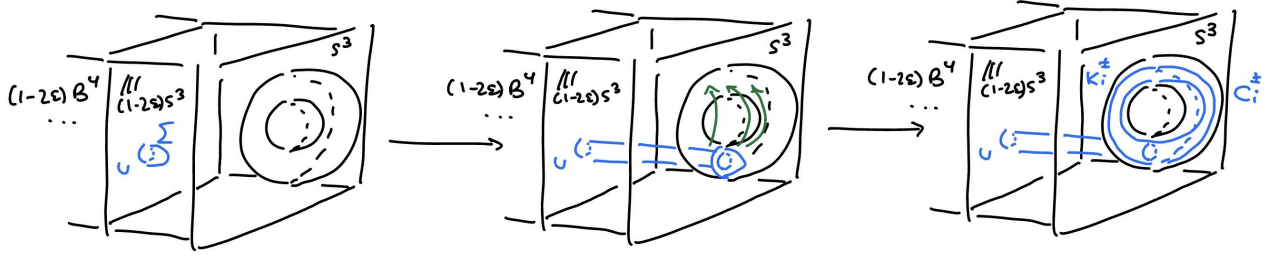


Figure 13: The 2-handle (drawn like a 3-dimensional 2-handle) is attached to S^3 . By pushing Σ into the 2-handle then performing the green isotopy, one retrieves $Z \cup C_i^\pm$

then one may exclusively work in the 2-handle $D^2 \times D^2$, where given a punctured annulus in $D^2 \times \partial D^2$ bounding $\{*, *'\} \times S^1 \sqcup U$ and two points $*, *' \in D^2$, we wish to isotope (rel U) the punctured annulus capped off at $\{*, *'\} \times S^1$ to a slice disk for U . We can perform this isotopy by sliding the capped off annulus along the 2-handle, as shown in Figure 13. Now plugging x^k into the $\text{KhR}_2(U)$ factor and applying Z induces the map ψ_i^k , so we find

$$[(\Sigma'_v, v \otimes x^k)] \cong [(\Sigma_{\psi_i^k(v)}, \psi_i^k(v))]. \quad (3.9)$$

Combining (3.8) and (3.9) shows that $\tilde{\Phi}(\psi_i^0(v)) = 0$, $\tilde{\Phi}(\psi_i^1(v)) = \tilde{\Phi}(v)$ as desired.

Let $\mathcal{C}_B(X, L)$ denote the full subcategory of $\mathcal{C}(X, L)$ of skeins whose input balls are strictly contained in B^4 . We now show Φ is an isomorphism by explicitly constructing its inverse Φ^{-1} alongside maps φ , φ' , ϕ as follows:

1. Consider a lasagna filling F on (X, L) with $[F] = \alpha \in H_2(X)$. Isotope all input balls out of the 2-handles to obtain a lasagna filling $\varphi(F)$ with skein in $\mathcal{C}_B(X, L)$. Here $\varphi(F)$ is well-defined up to isotopy rel. L .
2. Given any lasagna filling F' with skein in $\mathcal{C}_B(X, L)$, perturb F' to intersect each cocore transversely. Radially push outwards from the cocores again, until F' is only given by a collection of disks on each 2-handle. Consider the lasagna filling $\varphi'(F')$ obtained by enclosing the input balls of the resulting skein in a sufficiently large ball $(1-\epsilon)B^4$, and let $\phi(F')$ denote the class of the decoration of $\varphi'(F') \cap (1-\epsilon)S^3$ in $\text{KhR}_{2,[F']}(K)$.

Define $\Phi^{-1} = \phi \circ \varphi$. Since $[F] \in H_2(X)$ is invariant under this procedure, the number of positively and negatively oriented disks on the i^{th} handle differ by precisely $\alpha^+ + \alpha^-$, so we indeed have $\Phi^{-1}([F]) \in \text{KhR}_{2,\alpha}(K)$

Let us now check that Φ^{-1} is well-defined on $\mathcal{S}_0^2(X; L)$ and independent of the choices we made. Given any two lasagna fillings F_0, F_1 equivalent in $\mathcal{S}_0^2(X; L; \alpha)$, the skeins of $F = \varphi(F_0)$, $F' = \varphi(F_1)$ are objects in $\mathcal{C}_B(X, L)$ related by a sequence of skein morphisms in $\mathcal{C}(X, L)$ (regardless of which isotopies we chose to define F, F'). By repeatedly isotoping input balls away from the cocores and into B^4 , we can in fact relate F and F' by a sequence of skein morphisms in $\mathcal{C}_B(X, L)$. Thus it suffices to prove that any skein morphism in $\mathcal{C}_B(X, L)$ applied to a lasagna filling $F \in \mathcal{C}_B(X, L)$ does not change $\phi(F)$.

This roughly follows from Morse and transversality theory. Ball enclosure morphisms in $\mathcal{C}_B(X, L)$ clearly do not affect $\phi(F)$, so it suffices to consider skein isotopies $\{F_t\}_{t \in I}$ of lasagna fillings with skeins $\{\Sigma_t\}_{t \in I}$ satisfying $[\Sigma_t] = \alpha \in H_2^L(X)$. Let H_i be the i^{th} cocore. After adjusting the isotopy, we may assume the images $\cup_t \Sigma_t \times \{t\} \subset X \times I$ intersect $H_i \times I$ transversely. The intersections $\cup_t (\Sigma_t \cap H_i) \times \{t\}$ form a 1-dimensional submanifold S of $H_i \times I$. Under further generic conditions, we may assume:

1. Under the projection map $\pi : H_i \times I \rightarrow I$, S has finitely many critical points. Away from critical points, the isotopy $\{\Sigma_t\}$ induces an isotopy on $\varphi'(\Sigma_t)$, hence acting on $\Phi^{-1}(F_t)$ via the braid group action;
2. Around critical points, the isotopy $\{\Sigma_t\}$ is locally modelled near S such that $\varphi(\Sigma_t)$ precisely varies as per the push-in move from Figure 13 (i.e. a disk previously disjoint from the i^{th} 2-handle is pushed in, and thereafter bounds two oppositely oriented strands of K_i). Our previous discussion of the push-in move implies that $\Phi^{-1}(F_t) \in \text{KhR}_{2,\alpha}(K)$ is invariant under such moves. We can intuitively expect there to be only one local model around such critical points, as the 0-dimensional manifolds $\Sigma_t \cap H_i$ can only vary over t by isotopies, and by killing or generating pairs of oppositely oriented points.

In any case, we find that Φ^{-1} is well-defined. Now Φ, Φ^{-1} are inverses by construction, so we are done. $\square$

Remark 3.1. The 2-handle formula 3.5 shows that changing link orientations also only shifts the skein lasagna module by a bidegree. Explicitly, given a pair (X, L) , consider two classes $\alpha, \beta \in H_2(X, L)$ equivalent mod 2 such that $[\partial\alpha], [\partial\beta] \in H_1(L)$ are fundamental classes inducing different orientations on L (denoted $(L, \partial\alpha)$ and $(L, \partial\beta)$). Then for $* \in \{2, \text{Lee}\}$, we have (after tensoring with $\mathbb{Q}$ coefficients) [RW24, (16)]

$$\mathcal{S}_0^*(X; (L, \partial\beta); \beta) \cong \mathcal{S}_0(X; (L, \partial\alpha); \alpha) \left[\frac{\alpha^2 - \beta^2}{2} \right] \left\{ \frac{\beta^2 - \alpha^2}{2} \right\}. \quad (3.10)$$

4 Ren-Willis I: Lee Skein Lasagna

This section follows [RW24, Sections 4.1-5.2].

Recall that $\mathcal{S}_0^{\text{Lee}}(X; L)$ is the skein lasagna module of (X, L) associated to KhR_{Lee} . As usual $\mathcal{S}_0^{\text{Lee}}(X; L)$ inherits a homological $\mathbb{Z}$ -grading h and a quantum $\mathbb{Z}$ filtration q from the corresponding gradings in Lee homology. Here, due to the Khovanov-Rozansky renormalization (2.5), the filtration is now increasing with $q(0) = -\infty$. Note that since filtered maps do not necessarily preserve filtered degree, a nonzero element of $\mathcal{S}_0^{\text{Lee}}(X; L)$ may have filtration degree $-\infty$.

Previously, we found that the Lee deformation of Khovanov homology had a simple structure, and this had two important consequences:

1. The rigid structure of Lee homology allowed us to define the s -invariant, which is computable and reflects the functoriality of Khovanov / Lee (e.g. it is a concordance invariant).
2. We found a spectral sequence whose E^2 page is Khovanov homology and which converges to Lee homology. This let us bound Khovanov homology by computing the (much easier) Lee homology.

The first major task of [RW24] is to establish a theory of Lee skein lasagna. We will review this theory in Subsection 4.1 and introduce Ren-Willis' definition of the skein lasagna s -invariant in Subsection 4.1.2. Then in Subsection 4.2 we will give some tools for comparing the Lee and Khovanov skein lasagna modules.

4.1 Characterizing the Lee Skein Lasagna Module

Previously, we constructed a canonical basis for Lee homology using the orientations on the link. Our first major result in this section will similarly establish a basis for Lee skein lasagna parameterized by the possible homology classes of *double skeins*, i.e. pairs $(\Sigma, \mathfrak{D})$ where Σ is an (oriented) skein in (X, L) and $\mathfrak{D}$ is another orientation on Σ . This set of possible homology classes for double skeins is called *the set of double*

classes $H_2^{L,\times 2}(X)$ in (X, L) , and is defined explicitly as follows. Let $\{L_i\}$ be the set of components of L , and let $[L_i] \in H_1(L)$ be the fundamental class of L_i . Then

$$H_2^{L,\times 2}(X) := \left\{ (\alpha_+, \alpha_-) \in H_2(X, L)^2 \mid \partial \alpha_{\pm} = \sum_i \epsilon_{i,\pm} [L_i], \epsilon_{i,\pm} \in \{0, 1\}, \epsilon_{i,+} + \epsilon_{i,-} = 1 \right\}$$

Every double skein $(\Sigma, \mathfrak{D})$ defines a double class $([\Sigma_+], [\Sigma_-]) \in H_2^{L,\times 2}(X)$, by setting Σ_+ (resp. Σ_-) to be the components of Σ oriented as per $\mathfrak{D}$ (resp. opposite to $\mathfrak{D}$). Similarly, any class in $H_2^{L,\times 2}(X)$ is realised by a pair of oriented immersed surfaces (whose boundaries partition L) which generically intersect transversely at a finite collection of points. After replacing these points with open input balls, we have realised our given class with an oriented double skein.

Theorem 4.1 ([RW24], Theorem 4.1). *The double classes $(\alpha_+, \alpha_-) \in H_2^{L,\times 2}(X)$ give rise to canonical generators $x_{\alpha_+, \alpha_-} \in S_0^{\text{Lee}}(X; L)$ which form a basis of $S_0^{\text{Lee}}(X; L) \cong \mathbb{Q}^{H_2^{L,\times 2}(X)}$.*

The main idea behind this proof is that since Lee homology has such a nice characterization, the Lee skein morphism maps will retain enough information to allow us to classify S_0^{Lee} purely based on homological information (see also Theorem 3.1).

Proof (Sketch). We first construct canonical generators corresponding to each double skein. Given a lasagna filling (Σ, v) on (X, L) , obtain Σ° by adding small input balls to Σ intersecting Σ at unlinked unknots until every component of Σ intersects at least one input ball. This skein morphism gives a tensor product of unit maps $\text{KhR}_{\text{Lee}} \Sigma \rightarrow \text{KhR}_{\text{Lee}} \Sigma^\circ$; let v° be the image of v under this map. Note $[(\Sigma^\circ, v^\circ)] = [(\Sigma, v)]$ in $S_0^{\text{Lee}}(X; L)$. Then given a double skein $(\Sigma, \mathfrak{D})$, define the *canonical Lee lasagna filling* $x(\Sigma, \mathfrak{D})$ as

$$x(\Sigma, \mathfrak{D}) := \left(\Sigma^\circ, 2^{-n(\Sigma^\circ)} (-1)^{-n(\Sigma_-)} \bigotimes_i x_{\mathfrak{D}|_{K_i}} \right)$$

where $\{K_i\}$ is the set of input links of Σ° .

To prove Theorem 4.1, we will want to show that $[x(\Sigma, \mathfrak{D})]$ only depends on the double class of $(\Sigma, \mathfrak{D})$, and these classes together form a basis of $S_0^{\text{Lee}}(X; L)$. To this end, we will need to check the following two items (see [RW24, Lemmas 4.3, 4.4])

1. Any lasagna filling $[\Sigma, v]$ decomposes as a linear combination of lasagna fillings $[\Sigma, \otimes_i x_{\mathfrak{o}|_{K_i}}]$, where $\mathfrak{o}$ is an orientation on $\cup_i K_i$. For $S_0^{\text{Lee}}(X; L)$ to be generated by the $x(\Sigma, \mathfrak{D})$, we need to show that given any orientation $\mathfrak{o}$ not induced by an orientation of Σ , $[\Sigma, \otimes_i x_{\mathfrak{o}|_{K_i}}]$ vanishes in $S_0^{\text{Lee}}(X; L)$.

To see this, note that any such orientation has a component Σ_0 of Σ , and two components C_1, C_2 with incompatible orientations, lying on input links $K_{i_1} \subset B_{i_1}, K_{i_2} \subset B_{i_2}$ and on Σ_0 . Puncture Σ_0 by another small input ball B along an unknot U to get a new skein Σ_1 . Suppose the orientation $\mathfrak{u}_1$ on C_1 is compatible with the orientation $\mathfrak{u}$ on U ; then the orientation $\mathfrak{u}_2$ on C_2 is compatible with $\bar{\mathfrak{u}}$. Pick arcs γ_1, γ_2 on Σ_0 connecting U to C_1, C_2 . Then the Lee map given by the skein morphism induced by replacing $B_1 \cup B$ with $\nu(B_1 \cup \gamma_1 \cup B)$ maps $x_{\bar{\mathfrak{u}}} \otimes \otimes_i \mathfrak{o}|_{K_i}$ to 0 by (2.7) (this is essentially a pair of pants cobordism connecting C_1 and U , though possibly more complicated since C_1 is in a link). Similarly replacing $B_2 \cup B \mapsto \nu(B_2 \cup \gamma_2 \cup B)$ maps $x_{\mathfrak{u}} \otimes \otimes_i \mathfrak{o}|_{K_i} \rightarrow 0$, so

$$[\Sigma^\circ, \otimes \mathfrak{o}|_{K_i}] = [\Sigma_1, x_{\mathfrak{u}} \otimes x_{\mathfrak{o}|_{K_i}}] + [\Sigma_1, x_{\bar{\mathfrak{u}}} \otimes x_{\mathfrak{o}|_{K_i}}] = 0.$$

2. We will also need to show that skein morphisms interact very well with the canonical generators. In fact, given a skein morphism $[S] : \Sigma_1 \rightarrow \Sigma_2$, add small input balls to Σ_1 to adjust $[S]$ into a morphism $\Sigma_1^\circ \rightarrow \Sigma_2^\circ$. Any orientation $\mathfrak{D}$ on Σ_1 restricts to an orientation on Σ_2 . Then in fact $\text{KhR}_{\text{Lee}}([S])$ identifies $[x(\Sigma_1, \mathfrak{D})]$ with $[x(\Sigma_2, \mathfrak{D}|_{\Sigma_2})]$ in $S_0^{\text{Lee}}(X; L)$. (see [RW24, Lemma 4.4] for an explicit statement). One can show this by combining input 1 and (2.7).

With these two inputs we can prove Theorem 4.1 as follows. For any double class (α_+, α_-) let e_{α_+, α_-} be its corresponding generator in $H_2^{L, \times 2}(X)$. Then define $\mathcal{S}_0^{\text{Lee}}(X; L) \rightarrow \mathbb{Q}^{H_2^{L, \times 2}(X)}$ by mapping $[\Sigma, x(\Sigma, \mathfrak{D})] \rightarrow e_{[\Sigma_+, [\Sigma_-]}$ for every double skein $(\Sigma, \mathfrak{D})$. Using inputs 1, 2, one may prove this map is well-defined, and this map is surjective since every double class is represented by some double skein. For injectivity, repeat the proof of Theorem 3.1 to construct a framed cobordism rel L between any two double skeins $\Sigma_+ \cup \Sigma_-$ and $\Sigma'_+ \cup \Sigma'_-$, i.e. a pair of disjoint framed oriented 3-manifolds $Y_{\pm} : \Sigma_{\pm} \rightarrow \Sigma'_{\pm}$ outside the tubular neighborhood of a 1-complex in $\text{int}(X) \times I$. This produces a sequence of skein morphisms between $\Sigma_+ \cup \Sigma_-$ and $\Sigma'_+ \cup \Sigma'_-$ preserving the double class structure, which by Input 2 implies they define the same canonical Lee filling. This implies the map $\mathcal{S}_0^{\text{Lee}}(X; L) \rightarrow \mathbb{Q}^{H_2^{L, \times 2}(X)}$ is injective as desired. $\square$

From now on, given any double class (α_+, α_-) we will define x_{α_+, α_-} as $[x(\Sigma_+, \Sigma_-)]$ for any double skein $\Sigma = \Sigma_+ \cup \Sigma_-$ representing (α_+, α_-) . For later reference, our double classes satisfy the following grading properties, which can be proven with care using basic properties of Lee homology and the above ideas.

Theorem 4.2 ([RW24], Theorem 4.1). *The canonical generators x_{α_+, α_-} satisfy:*

1. x_{α_+, α_-} is homologically homogeneous with $h(x_{\alpha_+, \alpha_-}) = -2\alpha_+ \cdot \alpha_-$;
2. $[x_{\alpha_+, \alpha_-}] = \alpha_+ + \alpha_- \in H_2^L(X)$;
3. $q(x_{\alpha_+, \alpha_-}) = q(x_{\alpha_-, \alpha_+}) = \max(q(x_{\alpha_+, \alpha_-} \pm x_{\alpha_-, \alpha_+}))$. This also equals $\min(q(x_{\alpha_+, \alpha_-} \pm x_{\alpha_-, \alpha_+})) + 2$ if $\alpha_+ - \alpha_-$ is nonzero in $H_2(X, L; \mathbb{Q})$.

4.1.1 Lee Skein Lasagna Gluing

By applying Theorem 4.1 and the Lee cobordism map (2.7), we may retrieve a simple form for the gluing map induced by the following gluing setup (see also Figure 11). Consider a 4-manifold $X = X_1 \cup_Y X_2$ obtained by gluing X_1, X_2 along a part Y of their boundaries. Pick framed oriented tangles $T_0 \subset Y, T_1 \subset \partial X_1 - Y, T_2 \subset \partial X_2 - Y$ with common boundary on ∂Y . Let L, L_1, L_2 be the links $T_1 \cup T_2, T_0 \cup T_1, -T_0 \cup T_2$, and consider a surface S with boundary link L_2 . Interpret S as a lasagna filling by puncturing each component with input balls intersecting it at unknots, and assigning the coefficient 1 to each unknot. Then gluing skeins to S defines a gluing map (see also (3.3); here we evaluate S on the second factor there)

$$\mathcal{S}_0^{\text{Lee}}(X_2; S) : \mathcal{S}_0^{\text{Lee}}(X_1; L_1) \rightarrow \mathcal{S}_0^{\text{Lee}}(X; L),$$

with homological degree 0 and quantum filtration degree $-\chi(S) + \chi(T_0) - [S]^2$. Now, given classes $\alpha_i \in H_2^{L_i}(X_i)$ for $i \in \{1, 2\}$, let $\alpha_1 * \alpha_2$ denote their gluing in $H_2^L(X)$. Given double classes $(\alpha_{i,+}, \alpha_{i,-}) \in H_2^{L_i, \times 2}(X_i)$ (where $i \in \{1, 2\}$), we may glue them to a double class $(\alpha_{1,+} * \alpha_{2,+}, \alpha_{1,-} * \alpha_{2,-}) \in H_2^{L, \times 2}(X)$ if the pairs $(\partial\alpha_{1,+}, \partial\alpha_{2,+}), (\partial\alpha_{1,-}, \partial\alpha_{2,-})$ are *compatible*, i.e. agree on T_0 .

Theorem 4.3 ([RW24], Theorem 4.9). *The Lee skein lasagna gluing map sends*

$$\mathcal{S}_0^{\text{Lee}}(X_2; S)(x_{\alpha_+, \alpha_-}) = 2^{m(S, L_1)} \sum_{\mathfrak{D}} (-1)^{m(S_-, \mathfrak{D}), L_1, -} x_{\alpha_+ * [S_+(\mathfrak{D})]} x_{\alpha_- * [S_-(\mathfrak{D})]}$$

Here $\mathfrak{D}$ varies over all double skein structures on S compatible with (α_+, α_-) , and the decomposition $S = S_+ \cup S_-$ is given by $\mathfrak{D}$. The index $m(S, L_1)$ is an integer constant depending only on the topological type of (S, L_1) , which equals $n(\Sigma \cup S) - n(\Sigma)$ for any skein Σ on (X_1, L_1) .

One can prove Theorem 4.3 using Theorem 4.1 (with its two inputs) and the Lee cobordism map (2.7).

It turns out that concordances induce nice gluing maps. Specifically, consider a pair of framed links $K, K' \subset S^3$ bounding a framed concordance $C \subset S^3 \times I$. Then surgery on $S^3 \times I$ along C gives a homology cobordism W between the surgeries on S^3 along K and K' , which correspond respectively to $\partial X(K)$ and $\partial X(K')$. Notably gluing W gives an isomorphism $\phi_W : H_2(X(K)) \cong H_2(X(K'))$. We then have:

Theorem 4.4 ([RW24], Proposition 5.3). *In the above setup,*

$$S_0^{\text{Lee}}(W) : S_0^{\text{Lee}}(X(K)) \rightarrow S_0^{\text{Lee}}(X(K'))$$

is a $\mathbb{Z} \times H_2(X(K))$ -graded and quantum-filtered isomorphism (where ϕ_W identifies $H_2(X(K))$ and $H_2(X(K'))$). Notably, $s(X(K); L; \alpha) = s(X(K'); L'; \phi_W(\alpha))$ for all $\alpha \in H_2(X(K))$.

4.1.2 The Skein Lasagna s -invariant

Given our nice characterization of $S_0^{\text{Lee}}(X, L)$ 4.1, we may now expect the s -invariant to give useful information in the Lee context. We generalize its definition as follows:

Definition 4.1. *The lasagna s -invariant $s(X; L; \alpha_+, \alpha_-)$ of (X, L) at a double class $(\alpha_+, \alpha_-) \in H_2^{L, \times 2}(X)$ is given by*

$$s(X; L; \alpha_+, \alpha_-) := q(x_{\alpha_+, \alpha_-}) - 2\alpha_+ \cdot \alpha_-$$

For a class $\alpha \in H_2^L(X)$, we set $s(X; L; \alpha) := s(X; L; \alpha, 0)$. If $L = \emptyset$, we set $s(X; \alpha) := s(X; \emptyset; \alpha)$.

When X is a 2-handlebody, one may check that the isomorphism (3.10) induced by the orientation change $(L, \partial\beta) \rightarrow (L, \partial\alpha)$ maps a double class x_{β_+, β_-} to x_{α_+, α_-} , where $(\alpha_+, \alpha_-) \in H_2^{L, \times 2}$ is the unique double class with $\alpha_+ - \alpha_- = \beta_+ - \beta_-$. Hence by replacing $(L, \alpha) \rightarrow (L, \alpha_+ - \alpha_-)$ and $(\alpha_+, \alpha_-) \rightarrow (\alpha_+ - \alpha_-, 0)$ we may compute the double class s -invariants of (X, L) in terms of the lasagna s -invariants $s(X; L; \alpha)$.

Using Theorems 4.1 and 4.3 as well as the material in Subsection 3.2, one can readily prove many elementary properties of the lasagna s -invariant. For instance, we have $s(B^4; L; *) = s_{\text{KhR}}(L) + 1$ (where $*$ is the unique class in $H_2^L(B^4)$). Like the usual s -invariants, the lasagna s -invariants also satisfy genus bounds. For example, if $g(X; \alpha)$ denotes the minimum genus of a closed embedded surface Σ representing α , then we have $2g(X; \alpha) \geq s(X; \alpha) + \alpha^2$ [RW24, Corollary 4.17]. One can find more properties in [RW24, Subsection 4.3], though we will not need them.

4.2 Comparing Lee and Khovanov Skein Lasagna

Previously, we introduced the Lee spectral sequence as a tool for comparing $\text{Kh}(K)$ and $\text{Lee}(K)$. However, no spectral sequence yet exists linking the Khovanov and Lee skein lasagna modules. Nevertheless we have the following comparison result between them (by Ren-Willis) in the case of 2-handlebodies. In what follows, let $\text{gr}_*(V)$ denote the associated graded vector space of a filtered vector space V .

Theorem 4.5 ([RW24], Theorem 1.11). *Consider a 2-handlebody X . Then for any bidegree $(h, q) \in \mathbb{Z}^2$ and class $\alpha \in H_2^L(X)$ we have*

$$\text{rank}(S_{0,h,q}^2(X; L; \alpha)) \geq \dim(\text{gr}_q(S_{0,h}^{\text{Lee}}(X; L; \alpha))).$$

Proof sketch of Theorem 4.5. Work exclusively with $\mathbb{Q}$ coefficients (even for KhR_2). For elements a, b in the Lee complex, write $a \equiv^q b$ if $q(a) = q(b) > q(a - b)$ (i.e. their highest-graded parts are equal). Now, suppose X is the trace of a framed m -component link $K \subset S^3$. Let $K(r) := K(r + \alpha^+, r - \alpha^-) \cup L$ and let $S(r)$ be the product of symmetric groups $\prod_i S_{2r_i + |\alpha|}$. Recall by (3.7) and (3.6) that we may express S_0^* for $*$ $\in \{2, \text{Lee}\}$ as a colimit:

$$S_0^*(X; L; \alpha; \mathbb{Q}) \cong \text{colim}_{r \in \mathbb{Z}_{\geq 0}^m} \text{KhR}_*(K(r))^{S(r)} \{-|\alpha| - 2|r|\}. \quad (4.1)$$

Now, after some modifications, we may represent any basis of $\text{gr}_q S_{0,h}^{\text{Lee}}(X; L; \alpha)$ by a tuple of elements $a_1, \dots, a_n$ in the Lee complex $\text{CKhR}_{\text{Lee}}(K(r))\{-|\alpha| - 2r\}$, whose top q -graded parts are $c_1, \dots, c_n \in$

$\text{CKhR}_{\text{Lee}}(K(r)) \cong \text{CKhR}_2(K(r))$. Note $0 = d_{\text{Lee}} a_i \equiv^q d c_i$; since the c_i are q -homogeneous, they thus give (h, q) -graded classes in $\text{KhR}_2(K(r))\{-|\alpha| - 2|r|\}$. It now suffices to show any nontrivial linear combination c of the c_i does not vanish in $\mathcal{S}_0^2(X; L)$. Now, observe that the colimit (4.1) is filtered in the categorical sense (see [Wei94, Section 2.6]). Hence it suffices to show that c does not vanish under any colimit morphism $\text{KhR}_2(K(r))^{S(r)}\{-|\alpha| - 2|r|\} \rightarrow \text{KhR}_2(K(r'))^{S(r')}\{-|\alpha'| - 2|r'|\}$ [Wei94, Lemma 2.6.14], which is now an elementary exercise in graded algebra. $\square$

5 Ren-Willis II: Skein Lasagna for Positive Knot Traces

5.1 Outline

We now finally tackle the heart of this essay: Theorem 1.3. Theorem 1.3 will follow from Theorem 5.1, first proven by Ren-Willis, which constitutes one of the first nontrivial computations of the 2-handlebody formula. [Here's an outline of the ideas involved](#):

1. To start, the 2-handlebody formula is [hard](#), as you need to calculate the Khovanov homology of an infinite family of cables. Thus it will be advantageous to choose a setup which makes computation amenable. We will mainly work with a blackboard-framed [positive knot diagram](#) K . Such diagrams have the advantage that all their cables remain positive, and their resolutions are relatively easy to understand. We will briefly discuss the Khovanov homology of positive knots in Lemma 5.3.
2. It turns out that the skein lasagna module of $X(K)$ is not interesting when K is blackboard framed (for the interested reader, see [RW24, Theorem 1.4]). So we will need a tool to determine how adjusting the framing impacts the Khovanov-Rozansky homology of the resulting cables. This is the Comparison Lemma 5.1, and boils down to a computation of certain degrees in the Khovanov homology of a torus link. We will prove this lemma using an induction argument by Stošić [Sto09].
3. Using the Comparison Lemma and our results on Lee skein lasagna, we will be able to pin down the top homological degree of our skein lasagna module $\mathcal{S}_0^2(X(K), \emptyset)$. We will then use the structure of our cabled Khovanov Rozansky spaces as [symmetric group representations](#) to characterize their quantum filtration structures, and hence compute the s -invariants of $X(K)$.

5.2 The Main Calculation

From this section onwards work with $\mathbb{Q}$ coefficients for both Lee and KhR_2 . Given an oriented link L and integer vectors f, v , let L^f be L equipped with framing f , and let $L^f(v)$ be the v -cable of L^f (with all strands oriented as per L). Also for polynomials P, Q write $P \equiv^t Q$ (like in Subsection 4.2) if the highest t -degree terms of P, Q are identical. Our main computation is as follows:

Theorem 5.1 ([RW24], Theorem 1.13). *Consider a p -framed knot $K \subset S^3$ given by a knot diagram with N positive crossings and no negative crossings, where $p < 0$. Let $X = X(K^p)$ be its trace. For $\alpha \in \mathbb{Z}$, let $h(\alpha) = p\alpha^2/2$, and let $e(\alpha) := \#|h^{-1}(\{h(\alpha)\})|$ (which equals 1 if $\alpha = 0$ and 2 otherwise). Then for $\alpha \in \{-1, 0, 1\}$,*

$$\mathcal{S}_{0, > 0}^2(X; \alpha) = \mathcal{S}_{0, > 0}^{\text{Lee}}(X; \alpha) = 0; \quad (5.1)$$

$$\text{rank } \mathcal{S}_{0, 0}^2(X; \alpha) = \dim \mathcal{S}_{0, 0}^{\text{Lee}}(X; \alpha) = e(\alpha); \quad (5.2)$$

$$s(X; \alpha) = -s(-K^p(\alpha)) - w(K^p(\alpha)) - |\alpha| + 1 = \begin{cases} s(K) - p & \alpha \in \{1, -1\} \\ 1 & \alpha = 0 \end{cases} \quad (5.3)$$

Remark 5.1. This statement generalizes readily to link traces (see [RW24, Subsection 5.4]). Some main differences between the generalized variant and our knot variant are:

1. In the proof below, we will implicitly use that any cable of a blackboard-framed positive knot diagram has only positive intersections. This is false for general link traces, hence the statement of [RW24, Theorem 5.5] needs extra conditions;
2. In the general case, our results will compute $s(X; \alpha)$ for certain classes $\alpha \in H_2(X) \cong \mathbb{Z}^m$ which maximize $h(\alpha')$ across all $\alpha' \in \alpha + 2\mathbb{Z}^m$ (where the definition of h is also suitably generalized).

One may conclude Theorem 1.3 from Theorem 5.1 as follows:

Proof of Theorem 1.3. Recall the knots K_1, K_2 from Figure 1. As discussed in the introduction, it suffices to show that $X_1 := X_{-1}(K_1), X_2 := X_{-1}(K_2)$ are not diffeomorphic. To do this, first note that for any positive knot K and negative integer p , by applying (5.2), (5.3), and Theorems 4.1 and 4.2, we find

$$\text{rank}(\mathcal{S}_{0,0,q}^2(X_p(K); 1)) = \dim(\text{gr}_q(\mathcal{S}_{0,0}^{\text{Lee}}(X_p(K); 1))) = \begin{cases} 1 & q \in \{s(K) - p, s(K) - p - 2\}, \\ 0 & \text{else.} \end{cases} \quad (5.4)$$

Now observe that:

1. K_1 is clearly a positive knot, and one may compute $s(K_1) = 2$. Hence (5.4) gives

$$\text{rank}(\mathcal{S}_{0,0,q}^2(X_1; 1)) = \dim(\text{gr}_q(\mathcal{S}_{0,0}^{\text{Lee}}(X_1; 1))) = \begin{cases} 1 & q \in \{1, 3\}, \\ 0 & \text{else.} \end{cases}$$

2. On the other hand, K_2 is clearly ribbon slice. Hence by applying Theorem 4.5 to X_2 , Theorem 4.4 between K_2 and the unknot U , and (5.4) to U , we find

$$\text{rank}(\mathcal{S}_{0,0,q}^2(X_2; 1)) \geq \dim(\text{gr}_q(\mathcal{S}_{0,0}^{\text{Lee}}(X_2; 1))) = \begin{cases} 1 & q \in \{-1, 1\}, \\ 0 & \text{else.} \end{cases}$$

So $\text{rank}(\mathcal{S}_{0,0,-1}^2(X_1; 1)) < \text{rank}(\mathcal{S}_{0,0,-1}^2(X_2; 1))$, hence $\mathcal{S}_0^2(X_1; 1) \not\cong \mathcal{S}_0^2(X_2; 1)$ and thus X_1, X_2 are not diffeomorphic. □

Remark 5.2. In this proof it was not necessary to work with the Khovanov-Rozansky skein lasagna module, only the Lee skein lasagna module. Nevertheless we phrased results in terms of $\mathcal{S}_0^2(X)$ as this result also importantly gave nontrivial calculations of the Khovanov-Rozansky skein lasagna module.

5.3 The Comparison Lemma

We now state and prove the comparison lemma. Again, we will state the lemma for knots, but it generalizes immediately to links. Given a knot K , let $P(K)(t, q)$ and $P_{\text{KhR}}(K)(t, q)$ respectively denote the Poincaré polynomials of $\text{Kh}(K)$ and $\text{KhR}_2(K)$. Also define the *renormalized Khovanov-Rozansky homology* as $\widetilde{\text{KhR}}_2(K) := \text{KhR}_2(K)[w(K)/2]\{-w(K)/2\}$ (here the renormalization ensures $\widetilde{\text{KhR}}_2(K)$ is invariant under orientation reversal by 2.6) and denote its Poincaré Polynomial as $\tilde{P}_{\text{KhR}}(K)(t, q)$. Finally, given any two (Laurent) polynomials P, Q , write $P \leq Q$ if every coefficient in P is at most the corresponding coefficient in Q .

Lemma 5.1 ([RW24], Lemma 5.8). *Consider a knot $K \subset S^3$. For any nonnegative integer n and integers a, b with $a \leq b$, there exist polynomials $R_{n,n',b-a} \in \mathbb{Z}[t^{\pm 1/2}, q^{\pm 1/2}]$ independent of K for each $0 \leq n' \leq n$ such that*

$$\tilde{P}_{\text{KhR}}(K^a(n)) \leq \sum_{\substack{0 \leq n' \leq n \\ 2|n-n'}} R_{n,n',b-a} \tilde{P}_{\text{KhR}}(K^b(n')), \quad (5.5)$$

and the polynomials $R_{n,n',b-a}$ have highest t -degree term

$$R_{n,n',b-a}(t, q) \equiv^t t^{-(b-a)n'^2/2} q^{(b-a)(n'^2+2n')/2} \left(\sum_{r=0}^{(n-n')/2} \left(\binom{n}{\frac{n-n'}{2}-r} - \binom{n}{\frac{n-n'}{2}-r-1} \right) q^{2r} \right). \quad (5.6)$$

Moreover, if $n \geq 2$ and equality holds in (5.5) at the highest t -degree, then the annular creation map $\psi : \text{KhR}_2(K^b(n-2)) \otimes \text{KhR}_2(U) \rightarrow \text{KhR}_2(K^b(n))$ is injective in its highest nontrivial homological (h or t) degree.

Poincaré polynomials all have positive coefficients, so this is effectively just a relative bound on the highest t -degree term of $\tilde{P}_{\text{KhR}}(K^a(n))$. Thus in the proof of Lemma 5.1, we will only need to keep track of the highest t -degree terms in $R_{n,n',b-a}(t, q)$. By induction, it suffices to prove Lemma 5.1 in the case $b = a + 1$. In our proof we will also renormalize to Khovanov homology (where our normalized Poincaré polynomial is now $\tilde{P}(L) := (tq^3)^{-w(L)/2} P(L)$); Under this rephrasing, (5.5) becomes

$$\tilde{P}(K^1(n)) \leq \sum_{\substack{0 \leq n' \leq n \\ 2|n-n'}} R_{n,n'} \tilde{P}(K(n'))$$

where the polynomials $R_{n,n'} := R_{n,n',1}$ now must satisfy

$$R_{n,n'}(t, q) \equiv^t t^{-n'^2/2} q^{-(n'^2+2n')/2} \sum_{r=0}^{(n-n')/2} \left(\binom{n}{\frac{n-n'}{2}-r} - \binom{n}{\frac{n-n'}{2}-r-1} \right) q^{-2r}. \quad (5.7)$$

As previously discussed, we will prove Theorem 5.1 using the following induction argument of Stošić [Sto09]. The idea is as follows. Let $T_{m,n}$ be the (m, n) torus link. Give a knot diagram for K where its framing aligns with the blackboard framing. Then $K(n')$ is given by the $T_{n',0}$ -satellite of K , while $K(n)$ is given by the $T_{n,n}$ satellite (here, given a pattern $P \subset T^2$, we identify $T^2 \cong \partial \nu K$ by mapping the standard longitude on T^2 to the framing longitude on $\partial \nu K$). A satellite $P(K)$ may be given by constraining the nontrivial satellite data to a small arc on K , so Lemma 5.5 reduces to a computation of $\text{Kh}(T_{n,n})$. For this, we may induct on a generalized family of links $\{D_{n,m}^i\}$, which is closed under certain 0 and 1 resolutions, to be defined below. Inductively applying the Khovanov skein relation (2.4) and doing some combinatorics then finishes.

Here are the details. Consider the following families of links:

Definition 5.1 ([Sto09]). *Given $n, m \in \mathbb{Z}_{\geq 0}$ and $i \in [0, n-1]$, define $D_{n,m}^i$ as the braid closure of $(\sigma_1 \dots \sigma_{n-1})^m \sigma_1 \dots \sigma_i \in B_n$, regarded as a link in the solid torus $D^2 \times S^1$.*

Observe that $T_{n,n} = D_{n,n}^0$, $D_{n,m+1}^0 = D_{n,m}^{n-1}$, and $D_{0,m}^i = \emptyset$. Also, the braid definition of $D_{n,m}^i$ gives it a standard diagram, which we denote $\mathcal{D}_{n,m}^{i-1}$. For $k \in \{0, 1\}$, let $\mathcal{D}_{n,m}^{i-1,k}$ be the diagram obtained by k -resolving $D_{n,m}^i$ at its last letter σ_i . Then $\mathcal{D}_{n,m}^{i-1,0}$ depicts $D_{n,m}^{i-1}$ (see Figure 14). Now, define $E_{n,m}^{i-1}$ as the link depicted by $\mathcal{D}_{n,m}^{i-1,1}$ [Sto09]. To compute $\text{Kh}(T_{n,n}) = \text{Kh}(D_{n,n}^0)$, we will need the following knot-theoretic properties:

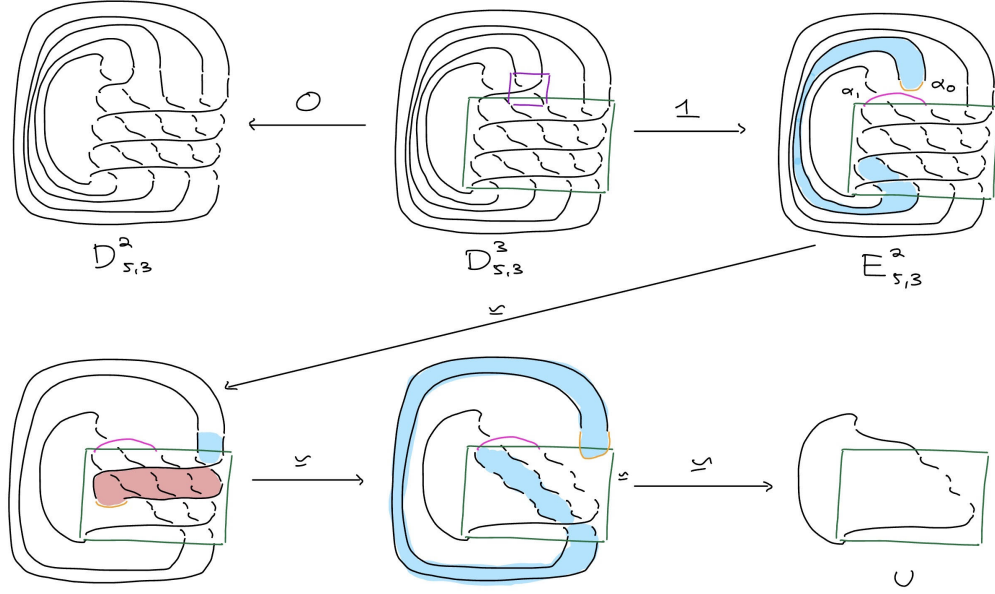


Figure 14: The two arrows in the top row show how 0, 1-resolving $D_{5,3}^3$ (at the purple intersection) respectively gives $D_{5,3}^2$, $E_{5,3}^2$. The other arrows demonstrate $E_{5,3}^2 \cong U$, by pushing the U -arc (orange) over (red) or under (blue) other arcs. Note that the green box always consists of $+1$ -rotated strands; for more general $E_{n,m}^{i-1}$, one may compute $D_{n_0,m_0}^{i_0}$ by counting which strands still connect to the green box.

Lemma 5.2 ([RW24], Lemma A.3). *For any $n, m \geq 0$, $i \geq 1$ there exist n_0, m_0, i_0 such that $E_{n,m}^{i-1}$ is isotopic in $D^2 \times S^1$ to either $D_{n_0,m_0}^{i_0}$ or $D_{n_0,m_0}^{i_0} \sqcup U$. Thus orient $\mathcal{D}_{n,m}^{i-1,1}$ as per $D_{n_0,m_0}^{i_0} (\sqcup U)$. Then we also have:*

1. $0 \leq n_0 < n$, $2|n - n_0$;
2. If $m < n$, then we may assume $m_0 < n_0$ or $n_0 = 0$;
3. $E_{n,n-1}^{n-2} \cong D_{n-2,n-3}^{n-3} \sqcup U$ and $E_{n,n-1}^i \cong D_{n-2,n-3}^i$ for $i < n-2$, where we set $D_{0,-1}^{-1} = D_{0,0}$;
4. For $m < n$, the number $\omega_{n,m}^i$ of negative crossings in $\mathcal{D}_{n,m}^{i-1,1}$ is at most $(n^2 - n_0^2)/2 - 1$, with equality iff $m = n - 1$.

The idea of the proof is best communicated by Figure 14, which demonstrates $E_{5,3}^2 \cong U$. In essence, 1-resolving the initial diagram for $D_{n,m}^i$ creates two arcs α_0, α_1 , one of which (WLOG α_0) connects two adjacent strands into a “U”. We can push this U through the solid torus $D^2 \times S^1$, deleting strands along the way, until it meets the other arc α_1 again. Carefully checking this operation proves the characterization of $E_{n,m}^{i-1}$ and items (1)-(3) (see e.g. [Tag13, Figures 15-20] for details). For (4), note that pushing the U under or over another arc deletes 1 positive and 1 negative crossing. So it suffices to estimate the number of arcs the U passes under / over, which reduces to elementary combinatorics (see [RW24, Lemma A.3]).

Proof of Lemma 5.1. View $D_{n,m}^i$, $E_{n,m}^i$ as satellite patterns, and give K by a knot diagram. We first give a recursive relation for $\text{Kh}(D_{n,m}^i(K))$ as follows. Consider a knot diagram for K where its framing aligns with the blackboard framing. Using this knot diagram, draw a diagram $\mathcal{D}_{n,m}^i(K)$ for $D_{n,m}^i(K)$ by inserting the pattern $D_{n,m}^i$ locally, and resolve $\mathcal{D}_{n,m}^i(K)$ at the last letter σ_i of $D_{n,m}^i$ to get diagrams

$\mathcal{D}_{n,m}^{i-1,0}(K), \mathcal{D}_{n,m}^{i-1,1}(K)$ respectively representing $D_{n,m}^{i-1}(K), E_{n,m}^{i-1}(K)$. One may count that the number of negative crossings in $\mathcal{D}_{n,m}^{i,1}(K)$ and $\mathcal{D}_{n,m}^i(K)$ differ by

$$\omega_{n,m}^i(K) := \frac{n^2 - n_0^2}{2} w(K) + \omega_{n,m}^i.$$

Thus, we obtain the following Khovanov skein exact sequence (see [Tur16, Theorem 1.1] for grading conventions):

$$\begin{array}{ccc} \mathrm{Kh}(D_{n,m}^i(K)) & \xrightarrow{\quad\quad\quad} & \mathrm{Kh}(D_{n,m}^{i-1}(K))\{1\} \\ & \nwarrow \quad \nearrow [1] & \\ & \mathrm{Kh}(E_{n,m}^{i-1}(K))[\omega_{n,m}^i(K) + 1]\{3\omega_{n,m}^i(K) + 2\} & \end{array}$$

where the arrow labelled $[1]$ is $(1, 0)$ -bigraded. Using Lemma 5.2, express $E_{n,m}^{i-1}$ as some $D_{n_0,m_0}^{i_0}$ disjoint union $\epsilon_{n,m}^i \in \{0, 1\}$ unknots. Since $P(U) = q + q^{-1}$, the exact triangle readily gives:

$$P(D_{n,m}^i(K)) \leq q \cdot P(D_{n,m}^{i-1}(K)) + (t^{\omega_{n,m}^i(K)+1} q^{3\omega_{n,m}^i(K)+2} (q + q^{-1})^{\epsilon_{n,m}^i} \cdot P(D_{n_0,m_0}^{i_0}(K))). \quad (5.8)$$

Now, define the polynomials $S_{n,n',K}$ as follows. Draw the graph G with vertices $\{(n, m, i) : 0 \leq i, m < n\} \cup \{(0, 0, 0)\} / \sim$, where we identify $(n, m, 0) \sim (n, m-1, n-1)$. Assign the following weighted directed edges to G :

1. Add an edge $(n, m, i) \rightarrow (n, m, i-1)$ of weight q ;
2. For each $E_{n,m}^{i-1} \cong D_{n_0,m_0}^{i_0} \sqcup U^{\epsilon_{n,m}^i}$, add a directed edge $(n, m, i) \rightarrow (n_0, m_0, i_0)$ of weight $t^{\omega_{n,m}^i(K)+1} q^{3\omega_{n,m}^i(K)+2} (q + q^{-1})^{\epsilon_{n,m}^i}$.

Let $S_{n,n',K}$ denote the sum of the weights of the paths from $(n, n, 0)$ to $(n', 0, 0)$ in G , where the weight of a path is the product of its edges' weights. Then by iteratively applying the exact triangle, we obtain:

$$P(K^1(n)) \leq \sum_{\substack{0 \leq n' \leq n \\ 2 \mid n-n'}} S_{n,n',K} P(K(n')).$$

To now compute $S_{n,n',K}$, observe that Lemma 5.2(4) implies that every path $(n, n, 0) \rightarrow (n', 0, 0)$ has t -degree at most $\frac{n^2 - n'^2}{2} (1 + w(K))$. A path achieves equality iff it only runs over edges of the form $(n, n-1, i) \rightarrow (n-2, n-3, i_0)$, where by Lemma 5.2(3) we know $i_0 = i-2$ if $i = n-1$ and $i_0 = i-1$ otherwise. Count all such paths using elementary combinatorics (while noting only the paths $(n, n-1, n-1) \rightarrow (n-2, n-3, n-3)$ give an unknot); after renormalizing, one precisely retrieves (5.7) as desired.

Finally we discuss the equality case. If (5.5) is an equality, then for each edge $(n, m, i) \rightarrow (n_0, m_0, i_0)$ within a path $(n, n, 0) \rightarrow (n', 0, 0)$, the exact triangle splits into a short exact sequence with middle term $\mathrm{Kh}(D_{n,m}^i(K))$. Notably the map

$$\mathrm{Kh}(E_{n,n-1}^{n-2}(K)) \rightarrow \mathrm{Kh}(D_{n,n-1}^{n-1}(K))$$

is injective. Now $E_{n,n-1}^{n-2}(K) \cong D_{n-2,n-3}^{n-3}(K) \sqcup U = K^1(n-2) \sqcup U$ while $D_{n,n-1}^{n-1}(K) \cong K^1(n)$. Since $E_{n,n-1}^{n-2}(K) \rightarrow \mathrm{Kh}(D_{n,n-1}^{n-1}(K))$ is given by a 1-resolution, we may readily check that it coincides with the annular creation map $\psi : K^1(n-2) \sqcup U \rightarrow K^1(n)$, as desired. $\square$

Remark 5.3. Implicit in the proof is that $S_{n,n',K}$ loses its dependence on K after renormalizing. We can see this *a priori* since renormalizing the exact triangle to $\widehat{Kh}$ eliminates its dependence on $\omega_{n,m}^i(K)$, and thus makes the corresponding graph G independent of K .

5.4 Proof of the Main Calculation

To prove Theorem 5.1 we will need the following important property of positive links diagrams, presented in [Kho03] with no proof. Given an oriented positive link diagram $(D, \mathfrak{o})$, let G_D be the graph whose vertex set consists of Seifert circles in the resolution $D_{\mathfrak{o}}$, and edge set corresponds to resolved crossings connecting these Seifert circles.

Lemma 5.3 ([Kho03], Proposition 8). *Given a positive link diagram D with G_D connected, we have $\dim \text{Kh}^{h=0}(D) = \dim \text{Lee}^{h=0}(D) = 2$.*

Proof. The resolution $D_{\mathfrak{o}}$ 0-resolves all crossings of D . Thus the Lee generators corresponding to $\mathfrak{o}, \bar{\mathfrak{o}}$ are the unique Lee generators with homological degree 0, hence $\dim \text{Kh}^{h=0}(D) \geq \dim \text{Lee}^{h=0}(D) = 2$. Now let $V = \mathbb{Z}[X]/X^2$ and let $C_{\mathfrak{o}} = V^{|\mathcal{V}_D|}$ denote the subgroup of the Khovanov complex corresponding to $\mathfrak{o}$. Each bit $\pi \in \{0, 1\}^{\mathcal{V}_D}$ corresponds to a pure tensor $x_{\pi} := \otimes_{v_i \in \mathcal{V}_D} X^{\pi(v_i)} \in C_{\mathfrak{o}}$. Now, each edge $\epsilon_{i,j} = (v_i, v_j) \in \mathcal{E}_D$ gives a component $d_{\epsilon_{i,j}}$ of the Khovanov differential d , given by changing the corresponding 0-resolution to a 1-resolution. This induces the multiplication map $d_{\epsilon_{i,j}}(x_{\pi}) = x_{\pi, \epsilon_{i,j}}$, where $x_{\pi, \epsilon_{i,j}} := (\otimes_{v_k \in \mathcal{V}_D \setminus \{v_i, v_j\}} X^{\pi(v_k)}) \otimes (X^{\pi(v_i) + \pi(v_j)})$.

Each edge $\epsilon_{i,j}$ induces an action $\sigma_{i,j}$ on $\{0, 1\}^{|\mathcal{V}_D|}$ given by swapping the two bit entries corresponding to v_i, v_j . Now suppose a linear combination $x = \sum_{\pi} \lambda_{\pi} x_{\pi}$ lives inside $\ker d$. WLOG suppose x is q -homogeneous, i.e. $|\pi|$ is constant for all bits with nonzero λ_{π} . Then the condition $d_{\epsilon_{i,j}} x = 0$ implies that $\lambda_{\pi} = 0$ whenever $\pi(v_i) = \pi(v_j) = 0$, and $\lambda_{\pi} = -\lambda_{\sigma_{i,j}(\pi)}$ whenever $(\pi(v_i), \pi(v_j)) = (0, 1)$. Since G_D is connected, these two conditions imply $x = 0$ unless π has at most one 0 entry. The case where π has no 0s contributes one generator to $\text{Kh}^{h=0}(D)$, and the conditions $\lambda_{\pi} = -\lambda_{\sigma_{i,j}(\pi)}$ ensure that only one other generator arises from the case where π has one 0. This finishes. $\square$

Proof of Theorem 5.1. Recall that our knot K is given a positive knot diagram $\mathcal{D}$ with N (positive) crossings. Hence the blackboard framing of K in $\mathcal{D}$ is also N . As before, let p denote the framing on K ; we may assume $p \leq w$, as otherwise we can add positive Reidemeister 1 twists to our knot diagram until this is true. We will only prove the $\alpha \geq 0$ case, as the $\alpha < 0$ case is analogous. Work with $\mathbb{Q}$ coefficients for both $\mathcal{S}_0^2$ and $\mathcal{S}_0^{\text{Lee}}$. As per (4.1), write the 2-handlebody formula as

$$\mathcal{S}_0^*(X; \alpha) \cong \text{colim}_{r \in \mathbb{Z}_{\geq 0}} \text{KhR}_*(K^p(\alpha + r, r))^{\mathcal{S}_{\alpha+2r}\{-|\alpha| - 2r\}} \quad (5.9)$$

for $* \in \{2, \text{Lee}\}$. We first prove (5.1) and half of (5.2). Let $\mathcal{D}_n$ denote the usual diagram for $K^N(n)$. Observe by the definition of Lee and KhR_{Lee} that the maximal t -degree in $P_{\text{KhR}}(K^N(n))$ is at most the number of negative crossings in $\mathcal{D}_n$, hence the maximal t -degree in $\tilde{P}_{\text{KhR}}(K^N(n))$ is at most half the number of crossings in $\mathcal{D}_n$, which is $n^2 N/2$. Now, we may compute $w(K^p(\alpha + r, r)) = p\alpha^2 = 2h(\alpha)$; thus the Comparison lemma (5.5) reads

$$t^{h(\alpha)} q^{-h(\alpha)} P_{\text{KhR}}(K^p(\alpha + r, r)) = \tilde{P}_{\text{KhR}}(K^p(\alpha + r, r)) \leq \sum_{\substack{0 \leq n' \leq n \\ 2|n-n'}} R_{n, n', N-p} \tilde{P}_{\text{KhR}}(K^N(n')). \quad (5.10)$$

Plugging in (5.6), we find that every term on the RHS (multiplied by $t^{-h(\alpha)}$) has t -degree at most $h(n') \leq h(\alpha) \leq 0$. This holds for all n' , so combined with Theorem 4.5 this proves (5.1).

Using Theorem 4.2, one may readily show $t = 0$ is achieved. Indeed, canonical generators of $\mathcal{S}_{0,0}^{\text{Lee}}(X; \alpha)$ correspond to pairs $(\alpha_+, \alpha_-) \in H_2(X)^2$ with $\alpha_+ \cdot \alpha_- = 0$, $\alpha_+ + \alpha_- = \alpha$. Such pairs correspond to classes $\alpha' \in H_2(X)$ with $\alpha'^2 = \alpha$ and $2|\alpha' - \alpha|$. Now the intersection pairing on $H_2(X)$ is precisely $2h(\cdot)$, so this proves $\dim \mathcal{S}_{0,0}^{\text{Lee}}(X; \alpha) = e(\alpha)$, which is half of (5.2).

Now, let $\tilde{P}_{\text{Lee}}(K^p(n))$ denote the renormalized Poincaré polynomial for $\text{gr}(\text{KhR}_{\text{Lee}}(K^p(n)))$, let $V_{r,*} := \text{KhR}_*^0(K^p(\alpha + r, r))^{\mathcal{S}_r\{-|\alpha| - 2|r\}}$ (where the 0 superscript denotes the $h = 0$ piece), and let the rank

polynomial P denote its sum of q -coefficients $\text{rank}(P)(t) := P(t, 1)$ (as a polynomial in t). We now claim the following structural results:

1. (5.10) is an equality in highest t -degree, and;
2. $\tilde{P}_{\text{Lee}}(K^p(n)) \equiv^t \tilde{P}_{\text{KhR}}(K^p(n))$.

Item (1) is especially useful, as then the Comparison Lemma 5.1 implies that the dotted annular creation maps ψ^1 are injective, which will help tremendously for (5.3). Note that the Khovanov-Rozansky Lee spectral sequence has terms $\{V_{r,2}^q\}$ on the E_2 page converging to $\{\text{gr}_q V_{r,\text{Lee}}\}$ on the E^∞ page. Thus it generally implies $\tilde{P}_{\text{Lee}}(K^p(n)) \leq \tilde{P}_{\text{KhR}}(K^p(n))$, and item (2) gives equalities $V_{r,2} = \text{gr}_q V_{r,\text{Lee}}$ (where gr_q denotes the associated (q) -graded vector space). Notably, this implies the rest of 5.2. Indeed, note $\mathcal{S}_{0,0}^*(X; \alpha) \cong \text{colim}_r V_{r,*}$ by definition; since filtered colimits of vector spaces are exact, we find that $\mathcal{S}_{0,0}^2(X; \alpha) = \text{colim}_r V_{r,2}$ is the subspace of finite-graded elements in $\text{colim}_r V_{r,\text{Lee}} = \mathcal{S}_{0,0}^{\text{Lee}}(X; \alpha)$. Together with Theorem 4.5, this proves $\dim \mathcal{S}_{0,0}^2(X; \alpha) = \dim \mathcal{S}_{0,0}^{\text{Lee}}(X; \alpha)$, finishing (5.2).

We may show items (1) and (2) combinatorially as follows. First we bound the rank of the max- t -degree term of the RHS of (5.10). Note each term $R_{n,n',w-p}\tilde{P}(K^w(n'))$ has t -degree $h(n')$; since $h(n') \leq h(\alpha)$ with equality at $n' = \alpha$, only the $n' = \alpha$ term contributes. Now when $\alpha \neq 0$, $K^N(\alpha)$ is a positive link, so Lemma 5.3 implies $\dim \text{KhR}_2^0(K^N(\alpha)) = 2$. Thus the rank of $R_{n,\alpha,w-p}\tilde{P}(K^w(\alpha))$ (hence also the RHS of (5.10)) in its maximum t -degree $h(\alpha)$ is

$$\epsilon(\alpha) \sum_{r=0}^{(n-\alpha)/2} \left(\binom{n}{\frac{n-\alpha}{2} - r} - \binom{n}{\frac{n-\alpha}{2} - r - 1} \right) = \epsilon(\alpha) \binom{n}{\frac{n-\alpha}{2}}.$$

To bound the LHS of (5.10) we compute $\tilde{P}_{\text{Lee}}(K^p(n))$. By Lemma 2.4(1), we have that $\text{KhR}_{\text{Lee}}(K^p(n))$ decomposes into S_n -subrepresentations

$$\text{KhR}_{\text{Lee}}(K^p(n)) = \bigoplus_{\substack{-n \leq n' \leq n, \\ 2|n-n'}} \text{KhR}_{\text{Lee},n'}(K^p(n))$$

where $\text{KhR}_{\text{Lee},n'}(K^p(n))$ is the span of the canonical generators corresponding to orientations of $K^p(n)$ with n' more positive than negative strands. Since $w(K^p((n+n')/2, (n-n')/2)) = h(n')$, only the factors $\text{KhR}_{\text{Lee},n'}(K^p(n))$ with $h(n') = h(\alpha)$ achieve the max t -degree $h(\alpha)$. Since there are $e(\alpha)$ equality cases, we find that the max- t -degree rank of $\tilde{P}_{\text{Lee}}(K^p(n))$ is again $\epsilon(\alpha) \binom{n}{\frac{n-\alpha}{2}}$. We generally have $\tilde{P}_{\text{Lee}}(K^p(n)) \leq \tilde{P}_{\text{KhR}}(K^p(n))$, hence we retrieve $\tilde{P}_{\text{Lee}}(K^p(n)) \equiv^t \tilde{P}_{\text{KhR}}(K^p(n))$ and $\equiv^t$ equality in (5.10), proving items (1) and (2).

We now compute $s(X; \alpha)$. Let the orientation on K induce an orientation $\mathfrak{o}_n$ on each $K^p(n)$ (by just taking parallel oriented copies); note $x_{\mathfrak{o}_\alpha}$ spans $\text{KhR}_{\text{Lee},\alpha}(K^p(\alpha, 0))$, so it is invariant under the S_α action. Now in $H_2^{\emptyset, \times 2}(X)$, the double class $(\alpha, 0)$ is represented by a double skein with only a positive-oriented component, given by α parallel copies of the core of the 2-handle in $X(K)$, with their collar boundaries pushed into B^4 to meet some $(1 - \epsilon)B^4$ along its boundary. Thus under the 2-handlebody formula (5.9), $x_{\alpha,0}$ maps to a scalar multiple of the class $[x_{\mathfrak{o}_\alpha}]$.

Now for $* \in \{2, \text{Lee}\}$, let ψ_* (resp. ψ_*^1) denote the corresponding (resp. dotted) annular creation maps. By our equalities $V_{r,2} = \text{gr}_q V_{r,\text{Lee}}$ (and the naturality of the Lee spectral sequence), we have that ψ_2^1 is the corresponding associated graded map of ψ_{Lee}^1 . Since the maps ψ_2 (hence ψ_2^1) are injective, the maps ψ_{Lee}^1 exactly increase q -degree by 2. Thus the maximum q -degree on $\text{KhR}_{\text{Lee},\alpha}(K(\alpha + r, r))$ is $q(x_{\mathfrak{o}_\alpha}) + 2r = s_{\text{KhR}}(K^p(\alpha)) + 2r + 1$. The grade-preserving property of ψ_{Lee}^1 is also [almost](#) enough to compute $s(X; \alpha)$, [except the symmetrization in the 2-handlebody formula may affect the \$q\$ -grading](#). To

resolve this we the following representation-theoretic argument, originating from [RW24, Theorem 5.5], [Ren24, Proposition 4.4], and [Ren24, Section 4.3].

Let $W_{\alpha+2r}$ be a vector space with one generator for each size α subset of $[1, \alpha + 2r]$. $S_{\alpha+2r}$ naturally acts on $W_{\alpha+2r}$, and by considering Lee canonical generators, we find that $W_r \cong \text{KhR}_{\text{Lee}, \alpha}(K(\alpha + r, r))$ as $S_{\alpha+2r}$ -representations. Now, for integers $a \geq b$, let (a, b) denote the irreducible S_{a+b} -representation corresponding to a Young tableaux with rows of length a, b . It is an elementary exercise in representation theory to show that $W_{\alpha+2r}$ decomposes as:

$$W_{\alpha+2r} \cong \bigoplus_{0 \leq s \leq r} (\alpha + 2r - s, s).$$

Now, recall $\text{KhR}_2(U)$ has canonical Lee generators $a = (1 + X)/2$, $b = (1 - X)/2$. Endow $\text{KhR}_2(U)$ with a $\mathbb{Z}/2$ -action via $\iota : a \mapsto -b$; then $\text{KhR}_2(U)$ splits into the trivial representation $1 = \text{span}(X)$ and the sign representation $\epsilon = \text{span}(1)$. We claim the annular creation map

$$\psi_{\text{Lee}} : \text{KhR}_{\text{Lee}}(K^p(\alpha + r, r) \sqcup U) \rightarrow \text{KhR}_{\text{Lee}}(K^p(\alpha + r + 1, r + 1))$$

is $S_{\alpha+2r} \times \mathbb{Z}/2$ -equivariant [Ren24, Proposition 4.4]. Indeed, one can check that the involution ι corresponds to the isotopy flipping U upside down. By checking commutation relations between this isotopy and the braid action isotopies, one can check the $S_{\alpha+2r} \times \mathbb{Z}/2$ -equivariance of ψ_{Lee} up to sign; these sign conventions are worked out in [Ren24, Proposition 4.4]. Now, one may compute the identities:

$$\text{Res}_{S_{d-2} \times \mathbb{Z}/2}^{S_d}(d - r + 1, r + 1) = \begin{cases} (\bigoplus_{i=-1}^1 (d - r + i, r - i) \otimes 1) \oplus ((d - r, r) \otimes \epsilon) & r < \frac{d+1}{2} \\ ((d - r + 1, r - 1) \otimes 1) \oplus ((d - r, r) \otimes \epsilon) & r = \frac{d+1}{2} \end{cases}$$

Thus $\psi_{\text{Lee}}^0 = \psi_{\text{Lee}}(\cdot, 1)$ maps $(\alpha + 2r - s, s) \subset \text{KhR}_{\text{Lee}}(K^p(\alpha + r, r))$ to the unique $(\alpha + 2r - s, s) \otimes \epsilon$ factor of $\text{KhR}_{\text{Lee}}(K^p(\alpha + r, r))$. Since $q(\epsilon) = -1$ (under the renormalization (2.5)), ψ_{Lee}^0 is $(0, 0)$ -bigraded.

Now, note that **all nonzero elements in each irreducible $S_{\alpha+2r}$ -subrepresentation $(\alpha + 2r - s, s)$ of $\text{KhR}_{\text{Lee}}(K^p(\alpha + r, r))$ has the same quantum degree** (which we will denote $q_r(\alpha + 2r - s, s)$). Thus $q_r(\alpha + 2r - s, s) = q_{r+1}(\alpha + 2r + 1 - s, s + 1)$. As discussed before, $\max_{s=0}^r (q_r(\alpha + 2r - s, s)) = s_{\text{KhR}}(K^p(\alpha)) + 2r + 1$. Now, the above implies $\max_{s=1}^r (q_r(\alpha + 2r - s, s)) = \max_{s=0}^r (q_r(\alpha + 2r - s, s)) - 2$. Thus the maximum q -degree $\max_{s=0}^r (q_r(\alpha + 2r - s, s))$ is attained by the trivial subrepresentation $(\alpha + 2r, 0)$, which is the unique subrepresentation to survive under the symmetrization $\text{KhR}_{\text{Lee}}^0(K^p(\alpha + r, r))^{S_{\alpha+2r}}$. $x_{\alpha,0}$ is represented by a scalar multiple of some (nonzero as ψ_{Lee}^1 is injective) element in $\text{KhR}_{\text{Lee}}^0(K^p(\alpha + r, r))^{S_{\alpha+2r}} \{-|\alpha| - 2r\}$, which has q -grading $s_{\text{KhR}}(K^p(\alpha)) + 2r + 1 - \alpha - 2r = s_{\text{KhR}}(K^p(\alpha)) - \alpha + 1$. This does not depend on r , so thus

$$s(X; \alpha) = s_{\text{KhR}}(K^p(\alpha)) - \alpha + 1 = -s(-K^p(\alpha)) - w(K^p(\alpha)) - |\alpha| + 1$$

by the grading shift (2.5), as desired! $\square$

6 Bonus: Exotica without Skein Lasagna

In a recent article, Nahm used Khovanov homology to establish an exotic pair of Mazur manifolds (contractible 4-manifolds with one 1-handle and one 2-handle) [Nah25]. A method for doing this was also sketched by Ren-Willis [RW24, Remark 6.9]; however Nahm's proof is special in that it is skein lasagna free, being based on Hayden-Sundberg's construction of exotic slice disks which uses only the functoriality of Khovanov homology up to sign [HS24]. We present this proof now.

Consider the family of knots $\{J_m\}_{m \in \mathbb{Z}}$ depicted in Figure 15, first described in [HS24, Figure 6]. Each knot J_m admits ribbon slice disks Σ_m, Σ'_m , where Σ_m (resp. Σ'_m) is obtained by adding two saddles in

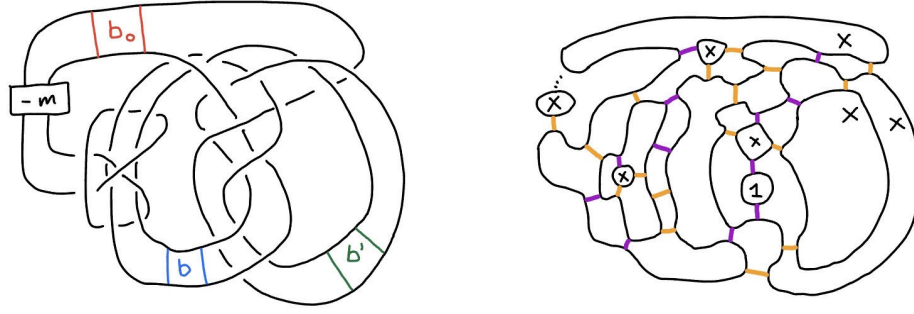


Figure 15: On the left is a diagram of J_m . On the right is a resolution π of J_m , with 0 resolutions shown in orange and 1 resolutions shown in purple. Each circle contributes one copy of $V = \mathbb{Z}[x]/x^2$ in the corresponding space C_π ; let α be the tensor product of the specified elements, one for each circle.

B^4 along the bands b, b_0 (resp. b', b_0) and capping off the remaining 3 unknots. In [HS24, Example 3.6], Hayden and Sundberg constructed the element $\alpha \in \text{Kh}(J_m)$ (see Figure 15), which by a computation does not vanish under the map $\text{Kh}_{\Sigma_m} : \text{Kh}(J_m) \rightarrow \text{Kh}(\emptyset) \cong \mathbb{Z}$ induced by Σ_m . On the other hand, the cobordism map $\text{Kh}_{\Sigma'_m}$ kills α , as the cobordism map induced by the saddle b' multiplies two X terms to give 0. Thus Σ_m and Σ'_m are not smoothly (ambiently) isotopic rel ∂ . Now Hayden and Sundberg proved that Σ_m and Σ'_m are topologically isotopic rel ∂ [HS24, Proposition 4.2]; thus the pairs (Σ_m, Σ'_m) became one of their first examples of *exotic slice disks* in B^4 .

Using these examples, Nahm showed that the 4-manifold pairs $(B^4 - \nu\Sigma_m, B^4 - \nu\Sigma'_m)$ are exotic as follows [Nah25]. By a known lemma [MWW22, Lemma 4.7], the non-existence of a smooth isotopy between Σ_m and Σ'_m implies that $B^4 - \nu\Sigma_m$ and $B^4 - \nu\Sigma'_m$ are not diffeomorphic rel ∂ . Thus it suffices to prove that the mapping class group of $B^4 - \nu\Sigma_m = S^3_0(J_m)$ (the 0-surgery on S^3 along J_m) is trivial, which one may do via hyperbolic 3-manifold methods (e.g. SnapPy computations, as used by Nahm). In fact, by blowing up the exotic disks Σ_m, Σ'_m and taking their complements in $\overline{\mathbb{CP}^2}$, Nahm proved the resulting pair of manifolds $\{Y_m, Y'_m\}$ form a pair of exotic *Mazur manifolds* [Nah25, Theorem 1.1].

In this article, we constructed Khovanov and Lee homology, and used it to prove the existence of exotic noncompact manifolds (Remark 2.1). We then showed Ren-Willis' analysis-free skein lasagna proof of the existence of a pair of exotic compact (simply connected) 4-manifolds. Just now, we discussed Nahm's skein-lasagna-free Khovanov proof of the existence of a pair of exotic compact contractible (Mazur) 4-manifolds using Khovanov homology. The next major goal is to use Khovanov homology to demonstrate an exotic pair of closed 4-manifolds. That task is left to the future.

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